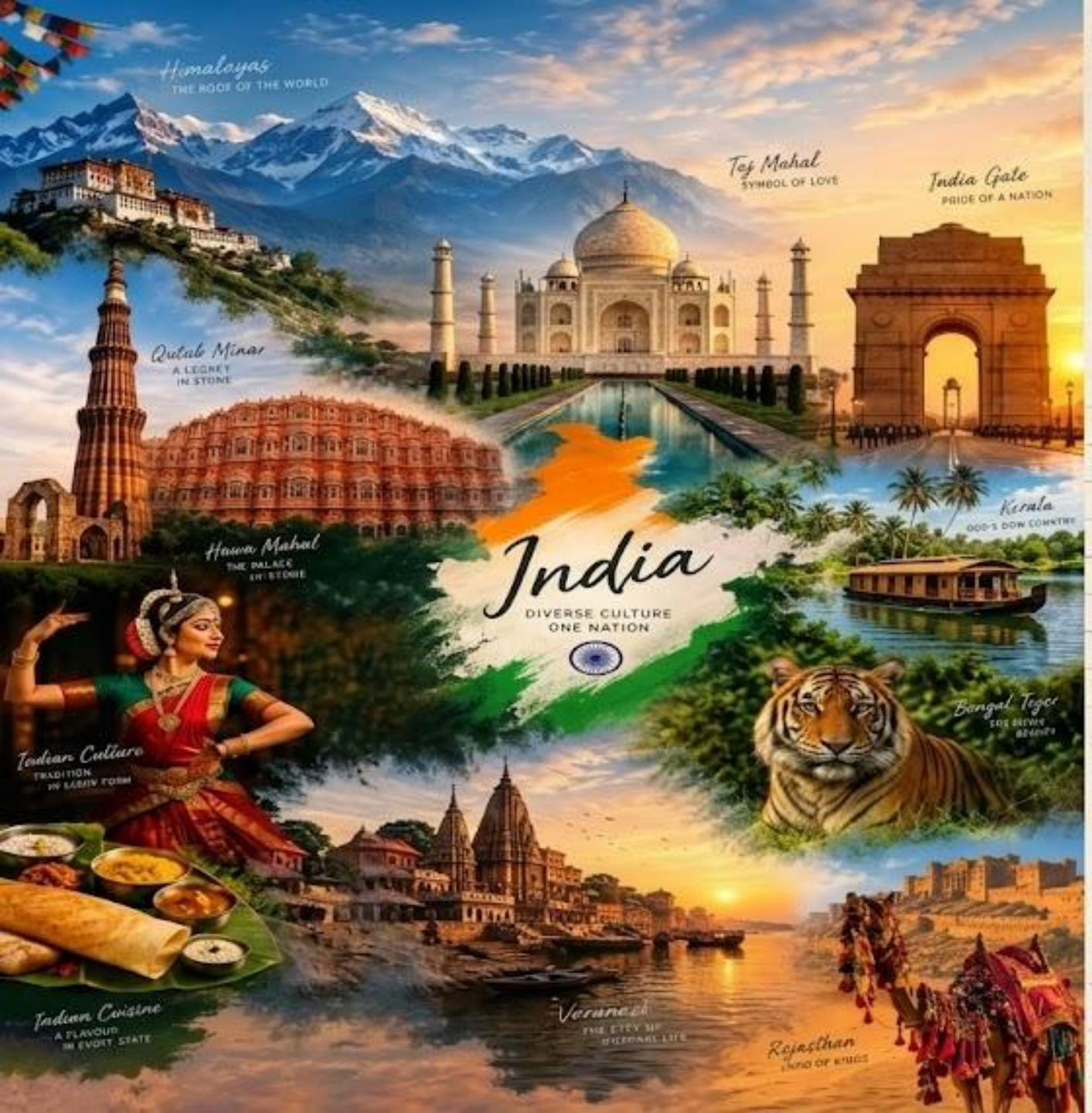


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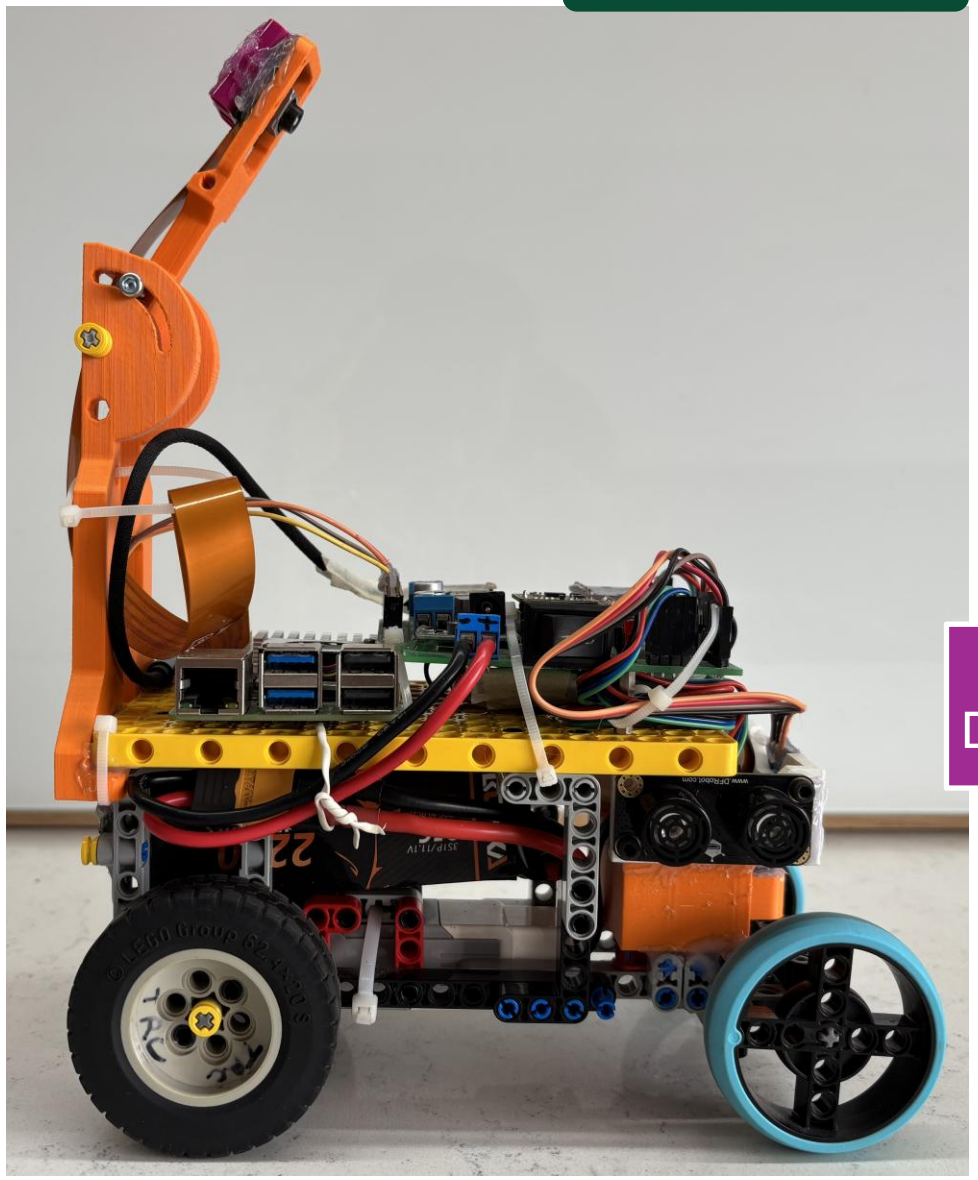


ROOTED IN INDIA
ENGINEERED FOR THE WORLD

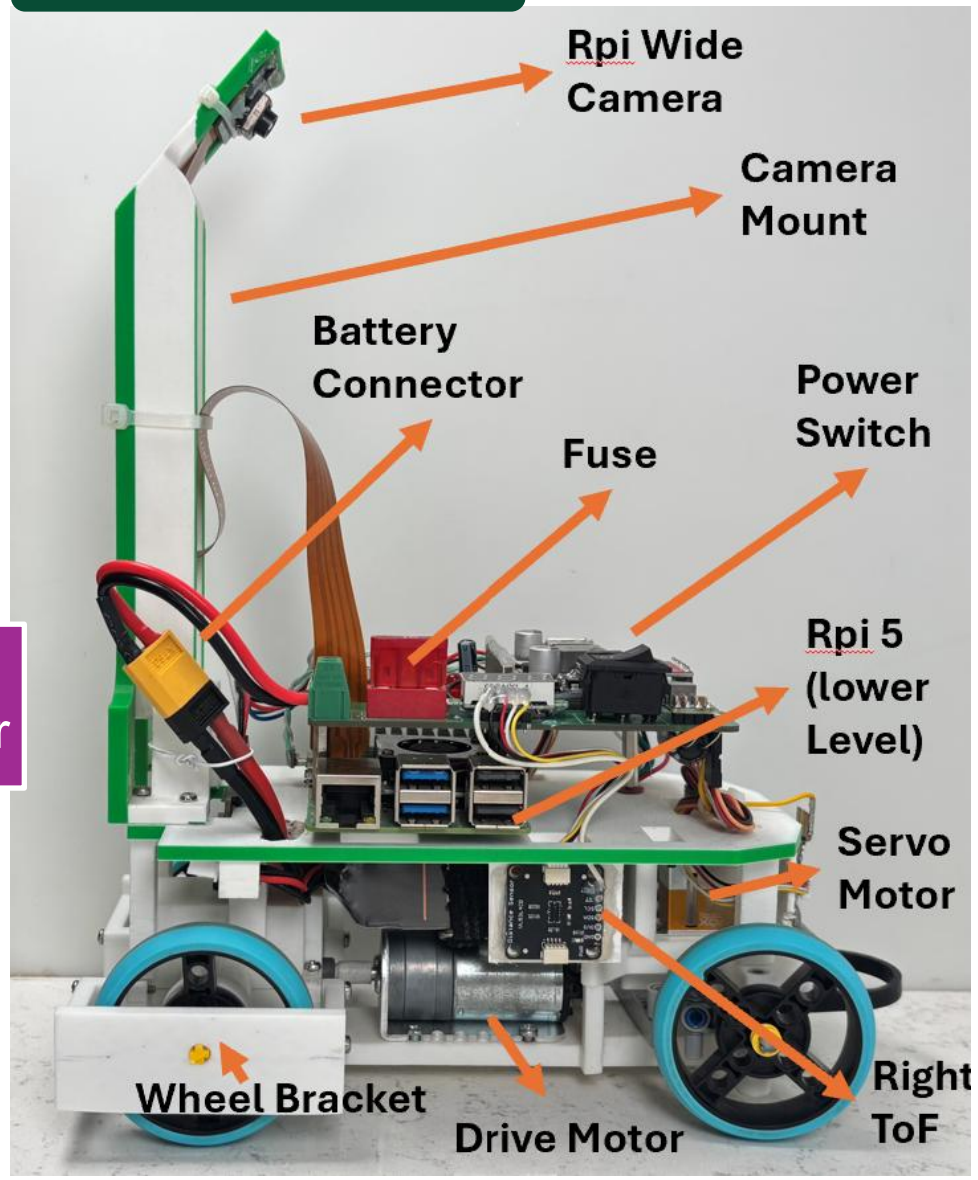
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WRO 2026

A journey of failures, discoveries, and relentless engineering on the quest for accuracy and speed.

2025 Singapore



2026 Puerto Rico



- 2x Faster
- 100% Accuracy
- Custom Chassis
- Metal Differential Gear
- Ackermann Steering
- Electrical Safety

Mobility & Mechanical design



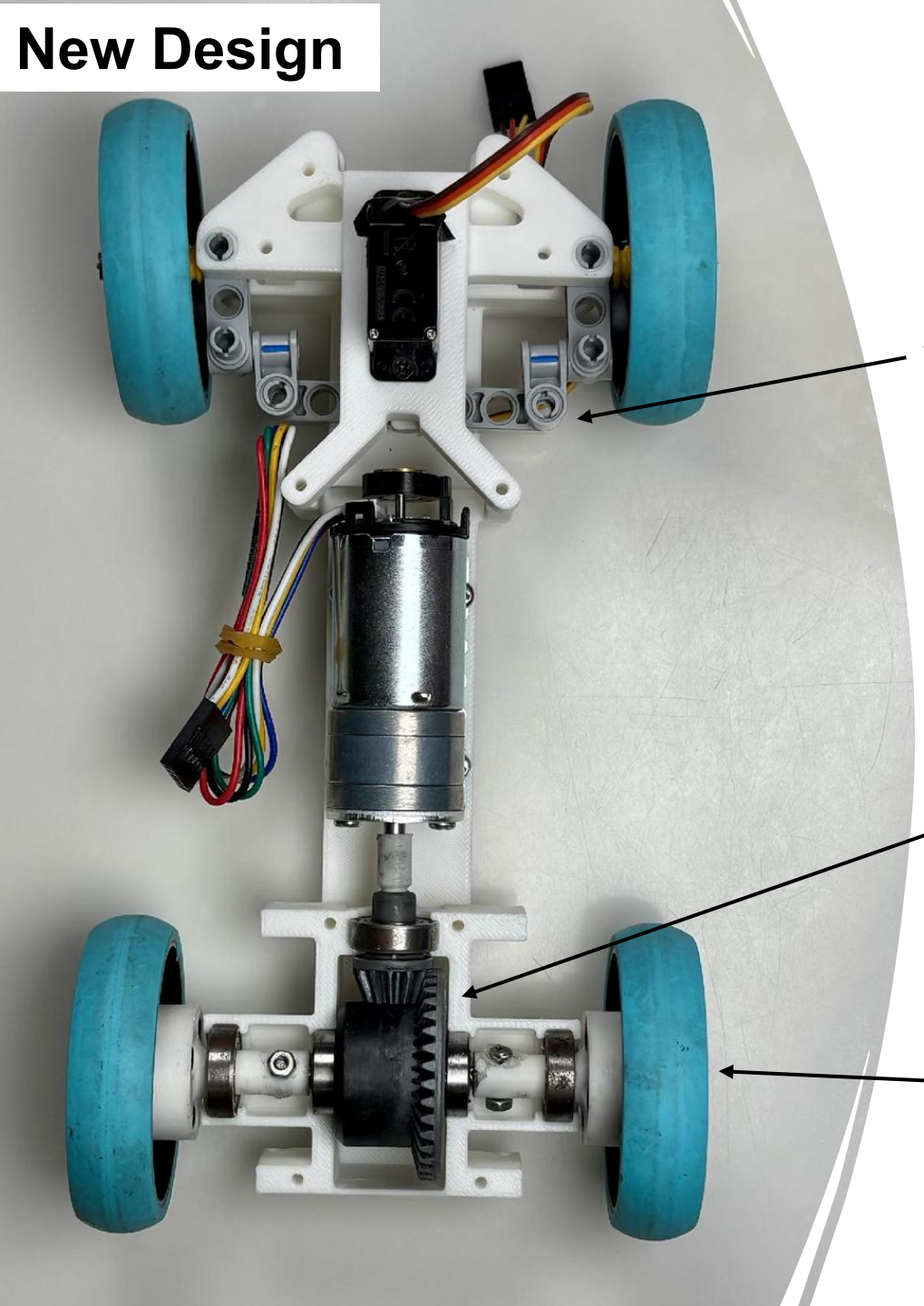
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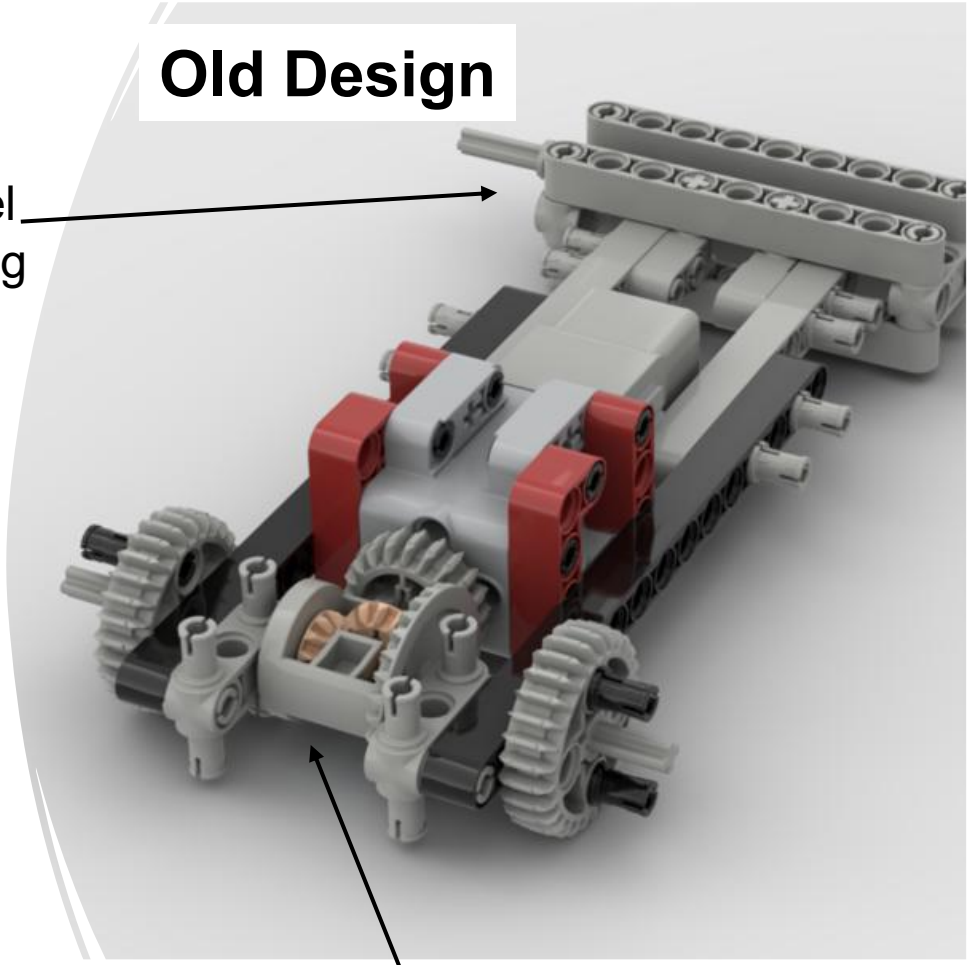
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Drive & Steering Choice

New Design

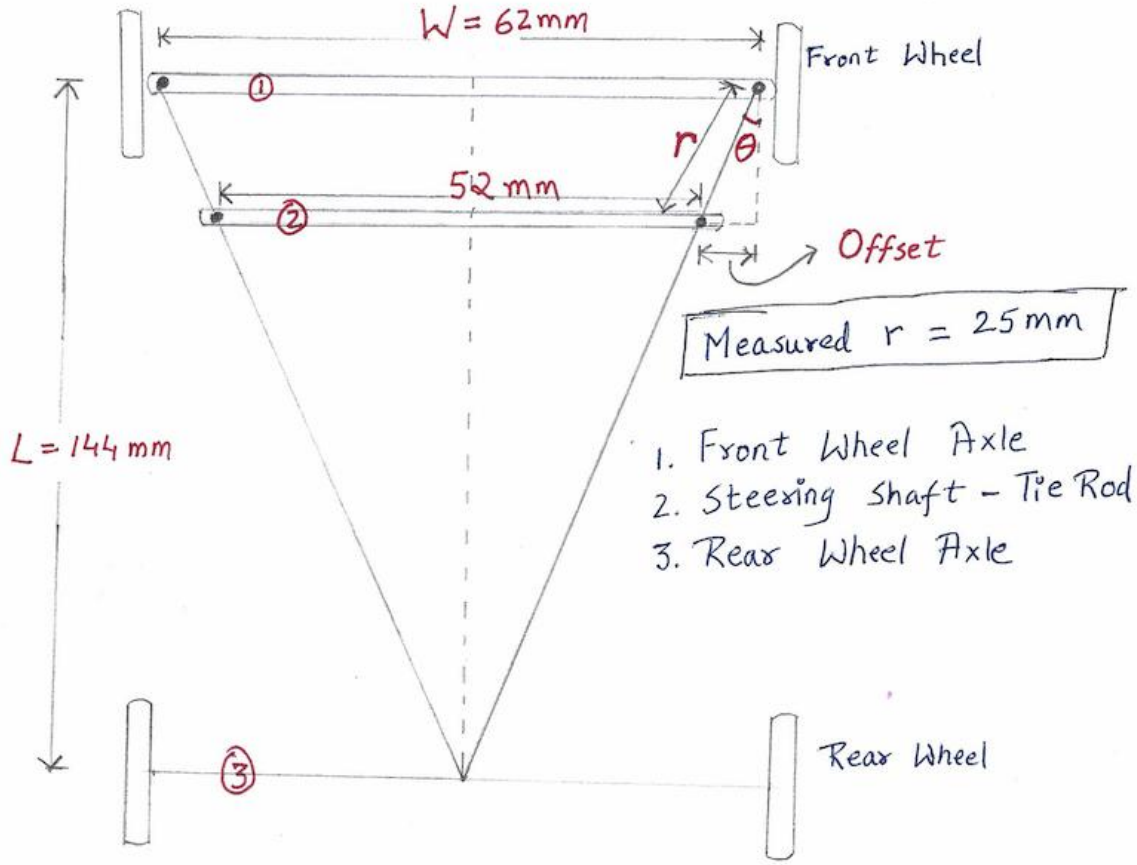


Old Design



Ackermann Calculations

Ackermann Steering Measurements



Ideal Ackermann Arm Angle

$$\theta = \tan^{-1}\left(\frac{W}{2L}\right)$$

where,

W = track width = 62 mm

L = Wheel base = 144 mm

$$\theta = \tan^{-1}\left(\frac{62}{2\times 144}\right)$$

$$= 12.15^\circ$$

Ackermann Steering Arm Relationship

$$\sin\theta = \frac{\text{Offset}}{r}$$

where,

r = steering arm length

$$\text{Offset} = \frac{\text{Front axle} - \text{Tie Rod}}{2}$$

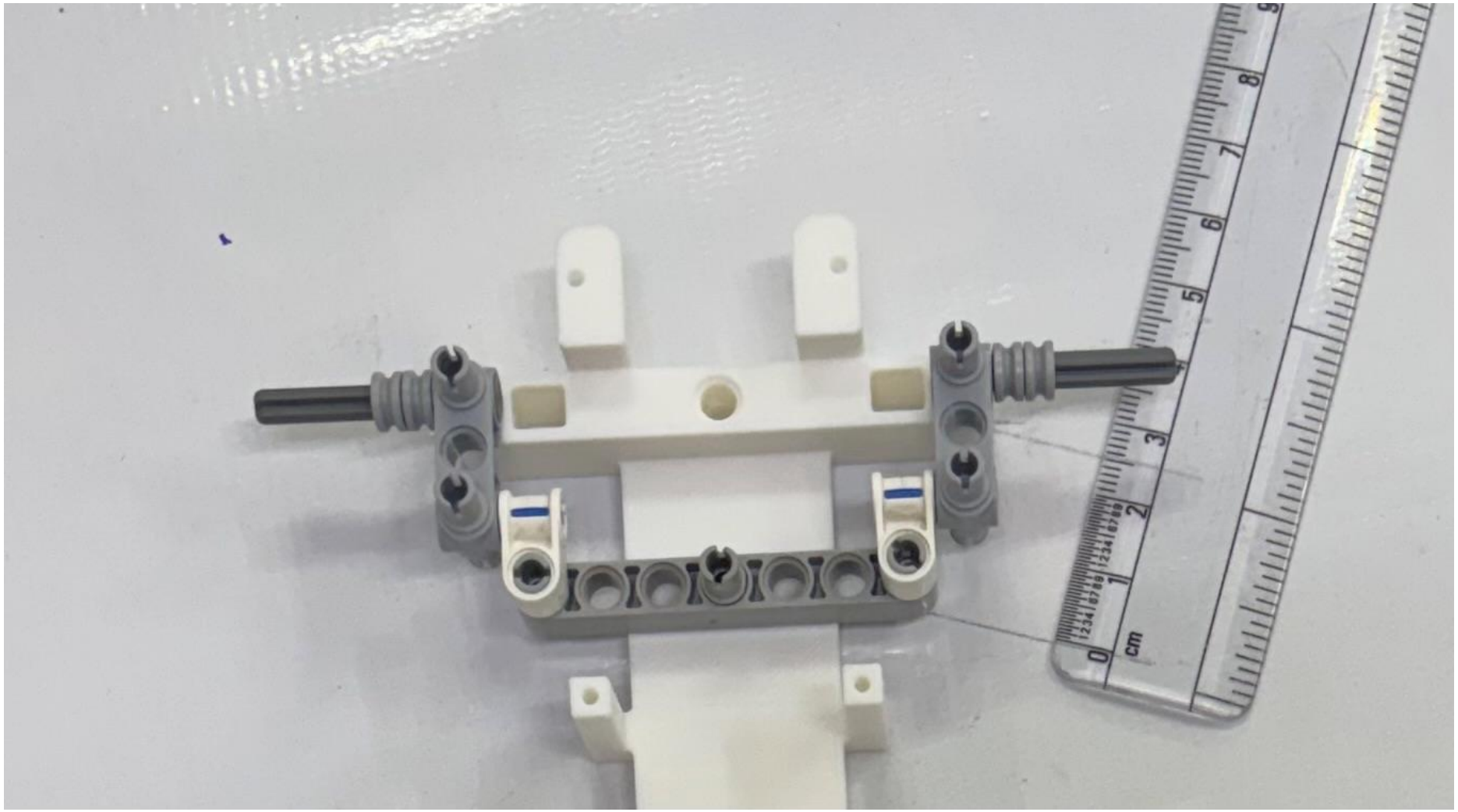
$$= \frac{62-52}{2} = 5 \text{ mm}$$

$$r = \frac{5}{\sin\theta} = \frac{5}{\sin(12.15)}$$

Calculated r = 23.8 mm

Calculated r ≅ Measured r

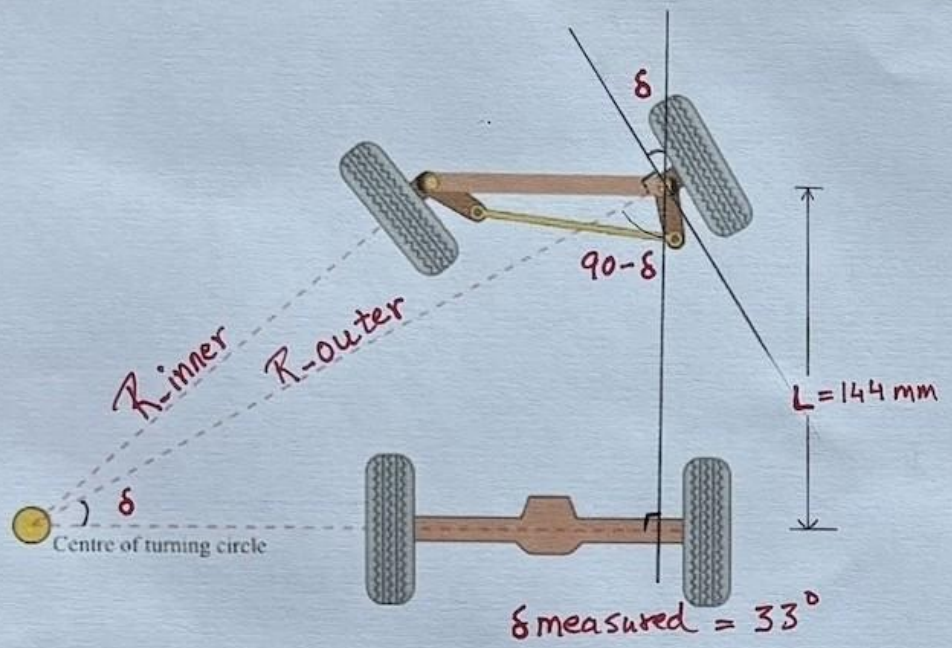
Ackermann Calculations



Calculation Backed Construction

Design Justification

Turning Radius Calculation

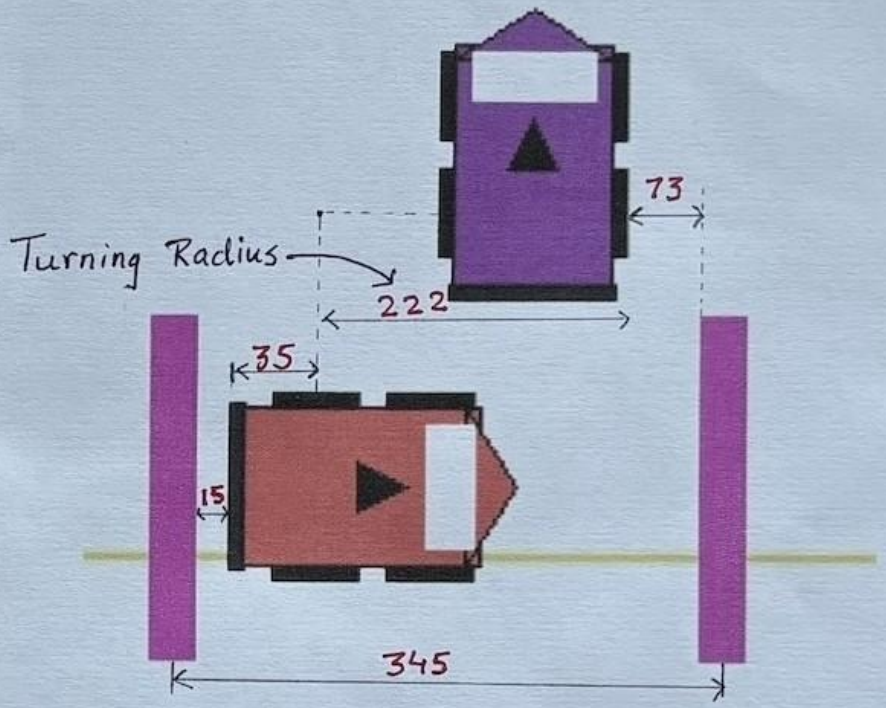


$$\tan(\delta) = \frac{L}{R_{outer}}$$

$$R_{outer} = \frac{L}{\tan(\delta)} = \frac{144}{\tan(33^\circ)}$$

Turning Radius = 222 mm

Parking Position Calculation using Turning Radius



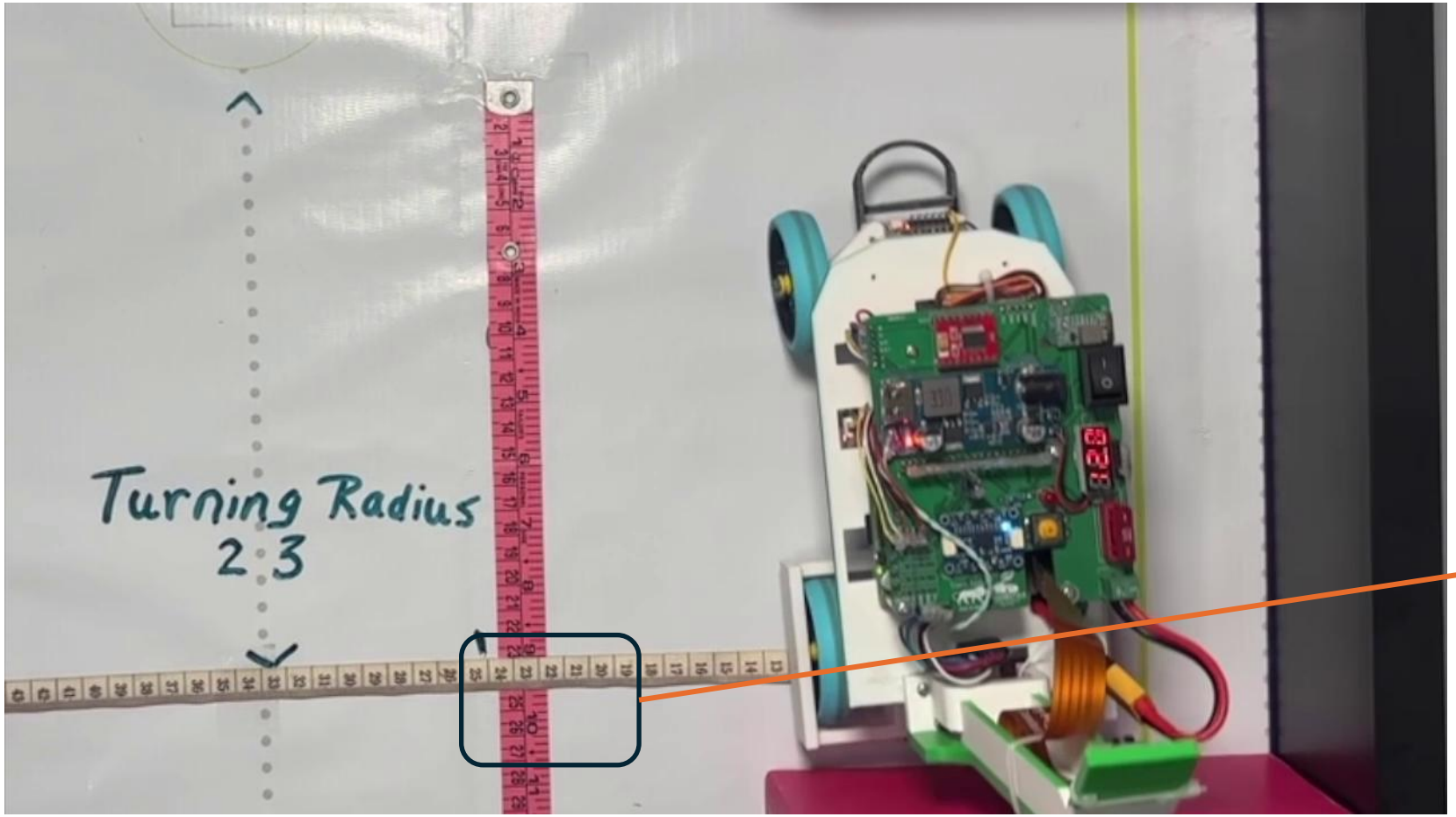
All lengths are in mm

Vehicle_length = 230 mm

Parking length = 1.5 × 230 = 345 mm

Turning Radius to Precise Parking Position

Actual Parking With Turning Radius Demo



Turning Radius ~ 230 mm

Drive Motor Selection

Specifications

Speed Torque Calculations

Stall Torque v/s Starting Torque

Step-1 : Breakaway Resistance force on flat ground - for both wheels.

F start = c or static x W (Includes stiction)

F start = 0.05 x 7.85 = 0.392 N

Step-2 : Torque Required at the ring gear

T req ring = F start x r w

T req ring = 0.392 x 0.028 = 0.0110 N.m

= 0.112 kg.cm

Step-3 : Referring to gearbox output

T req gb = T req ring / n ext = 0.112 / 2.923

= 0.0383 kg.cm

Step-4 : Comparision to available stall torque

Margin A = T stall gb / T req. gb = 0.71 / 0.0383

= 18.5x or 18.5 times

∴ The robot can easily start from a dead stop with the current setup.

Available torque is 18x more than Starting Torque

Running Torque v/s Continuous Torque

Step 1: Rolling Resistance experienced by the drivetrain (both wheels)

$F_{roll} = C_{rr} \times W$

$= 0.03 \times 7.85 = 0.236\ N$

Step 2: Torque required at the ring gear

$T_{req\ ring} = F_{roll} \times r_w$

$= 0.236 \times 0.028 = 0.0066\ N.m$

$= 0.0673\ kg\ cm$

Step 3: Referring to actual gearbox output

$T_{req\ gb} = T_{req\ ring} / G_{ext} = 0.0673 / 2.923$

$= 0.0230\ kg.cm$

Step 4: Continous Torque Rating Available

$T_{cont\ gb} = 0.25 \times T_{stall\ gb} = 0.25 \times 0.71$

$= 0.1775\ kg.cm$

Step 5: Compare

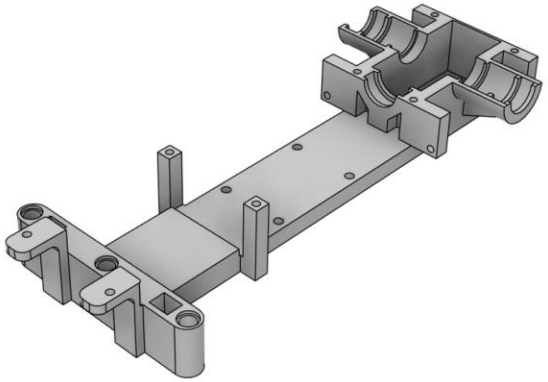
Margin Cl = $T_{cont\ gb} / T_{req\ gb} = 0.1775 / 0.0230$

= 7.72 or 7.7 times

∴ Continous running uses only 13% of the recommended continous torque rating, so the motor is running under very light load.

Running Torque is only 13 % of the Motor's Continuous Torque rating

3D custom Printed Parts



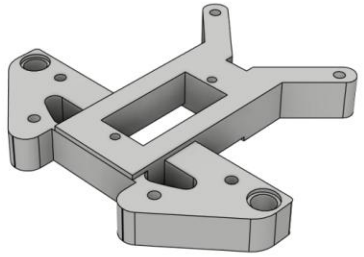
Bottom Chassis



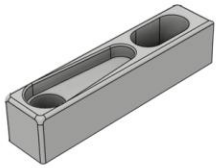
Bearing To Wheel Shaft



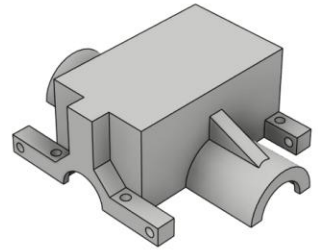
Motor Shaft



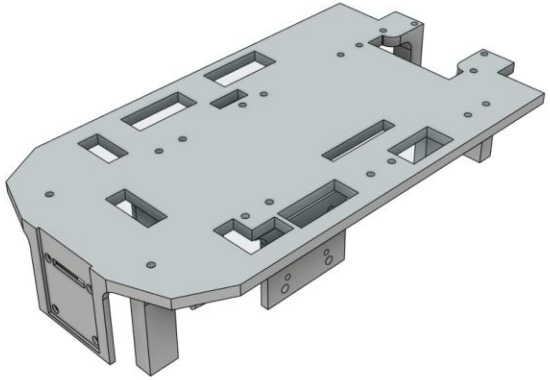
Servo Mount



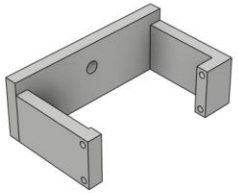
Variable Servo Horn



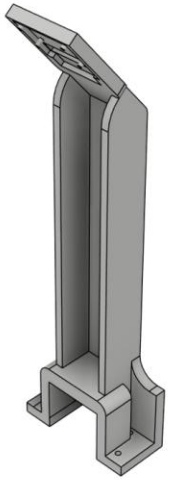
Differential Mount Top



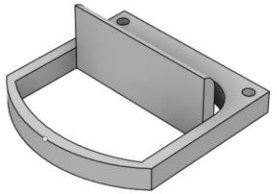
Top Chassis



Wheel Bracket

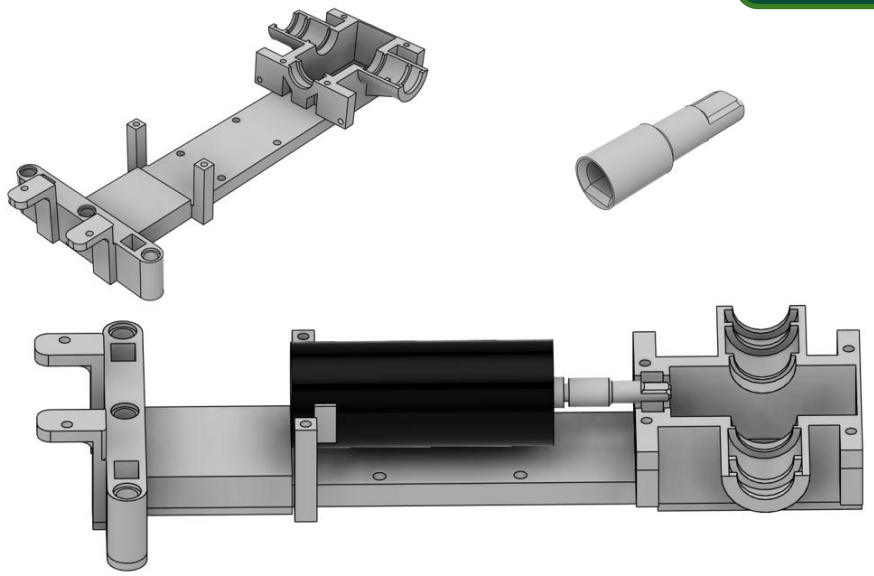


Camera Mount

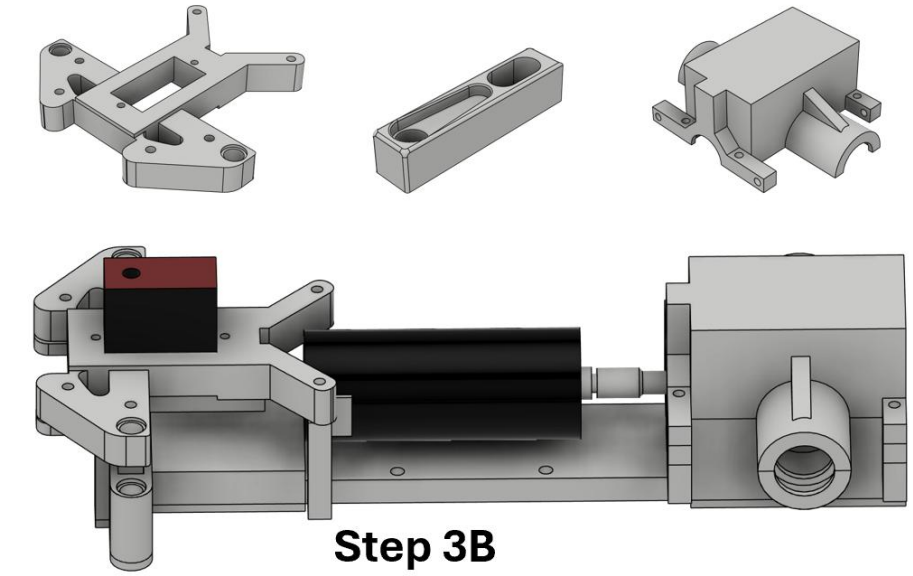


Bumper

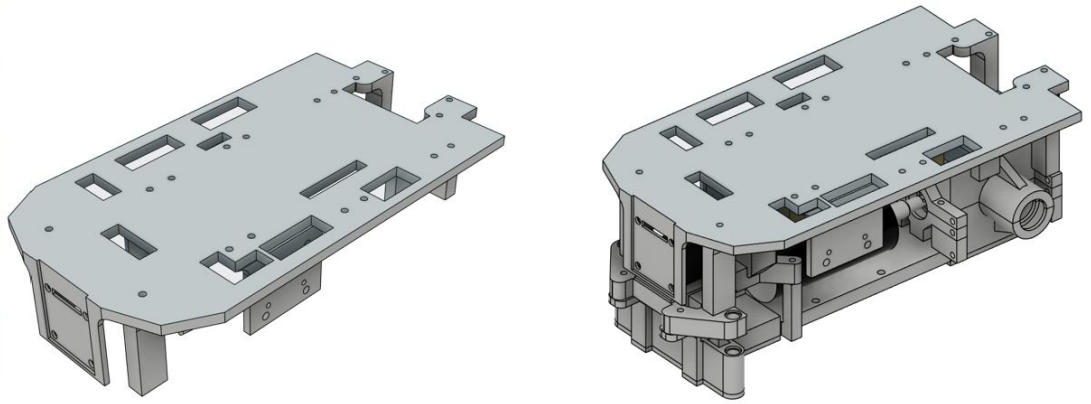
Chassis Mounting



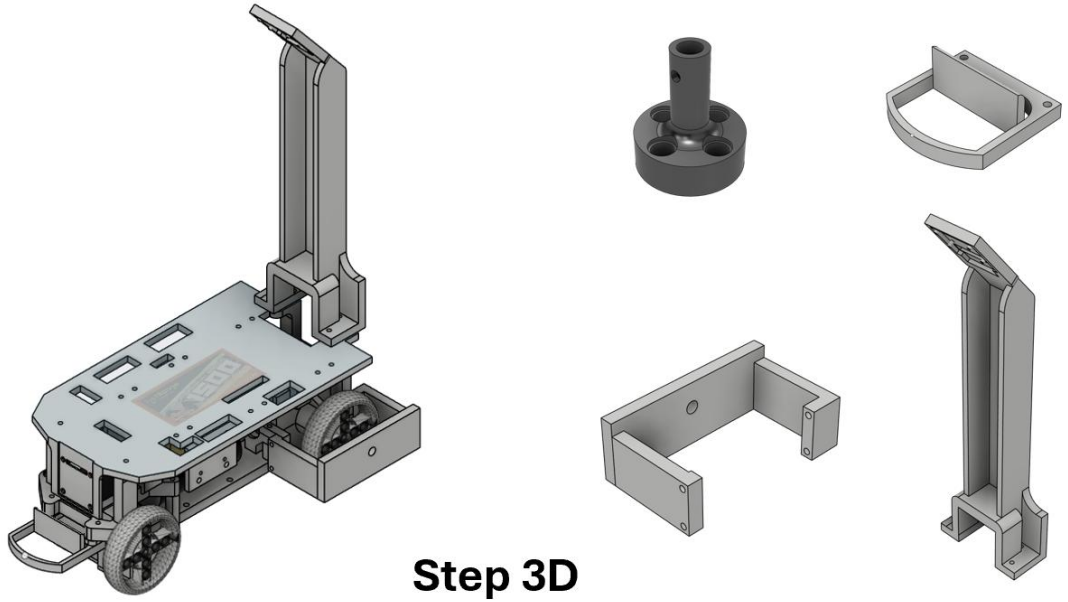
Step 3A



Step 3B



Step 3C



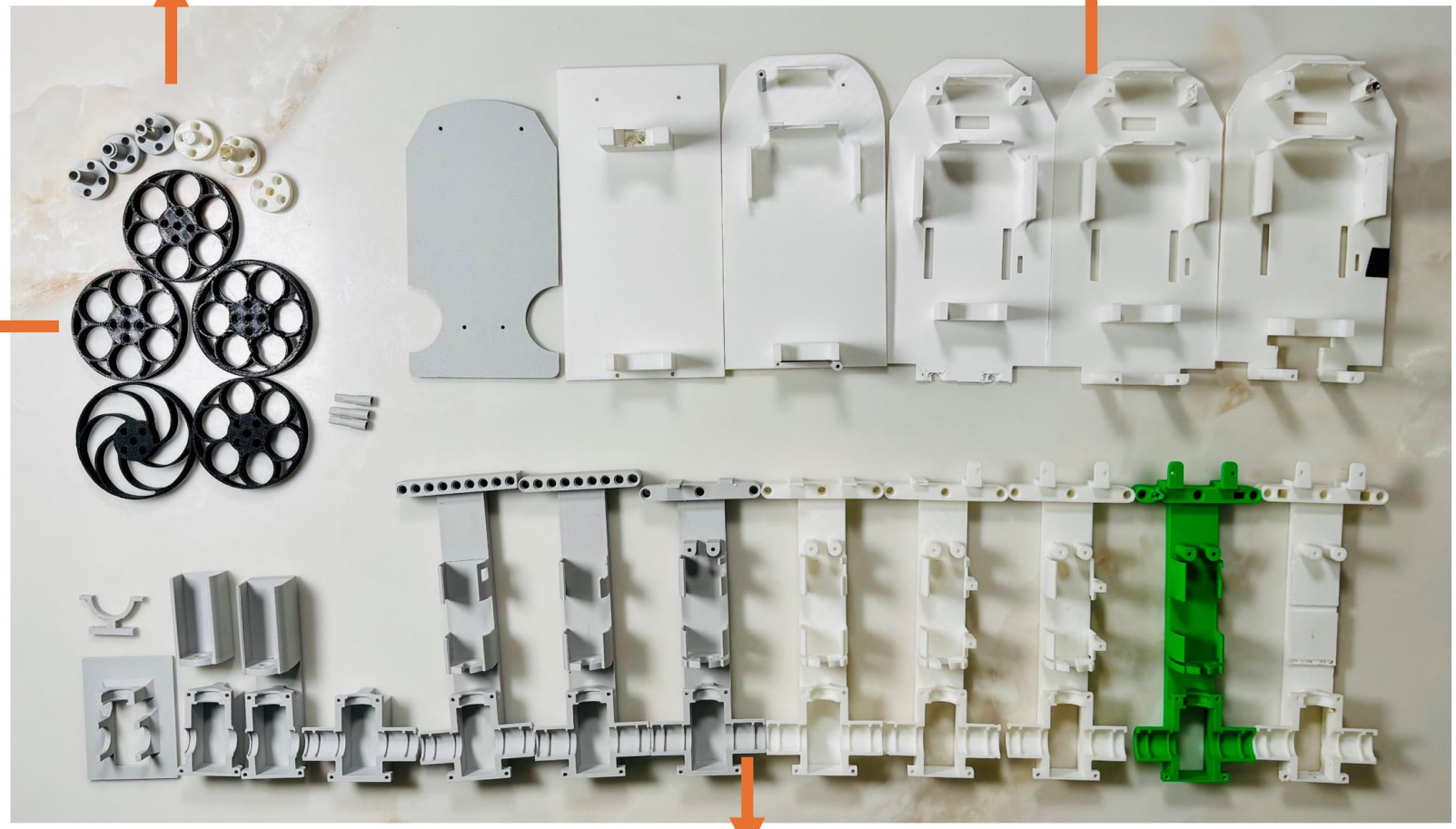
Step 3D

Chassis Iterations

Gear Wheel Connectors – evolved with Base chassis

Top Chassis Iterations V 1-6 sensor placement, wiring holes, attachment to bottom chassis

Custom Wheels- Failed attempt

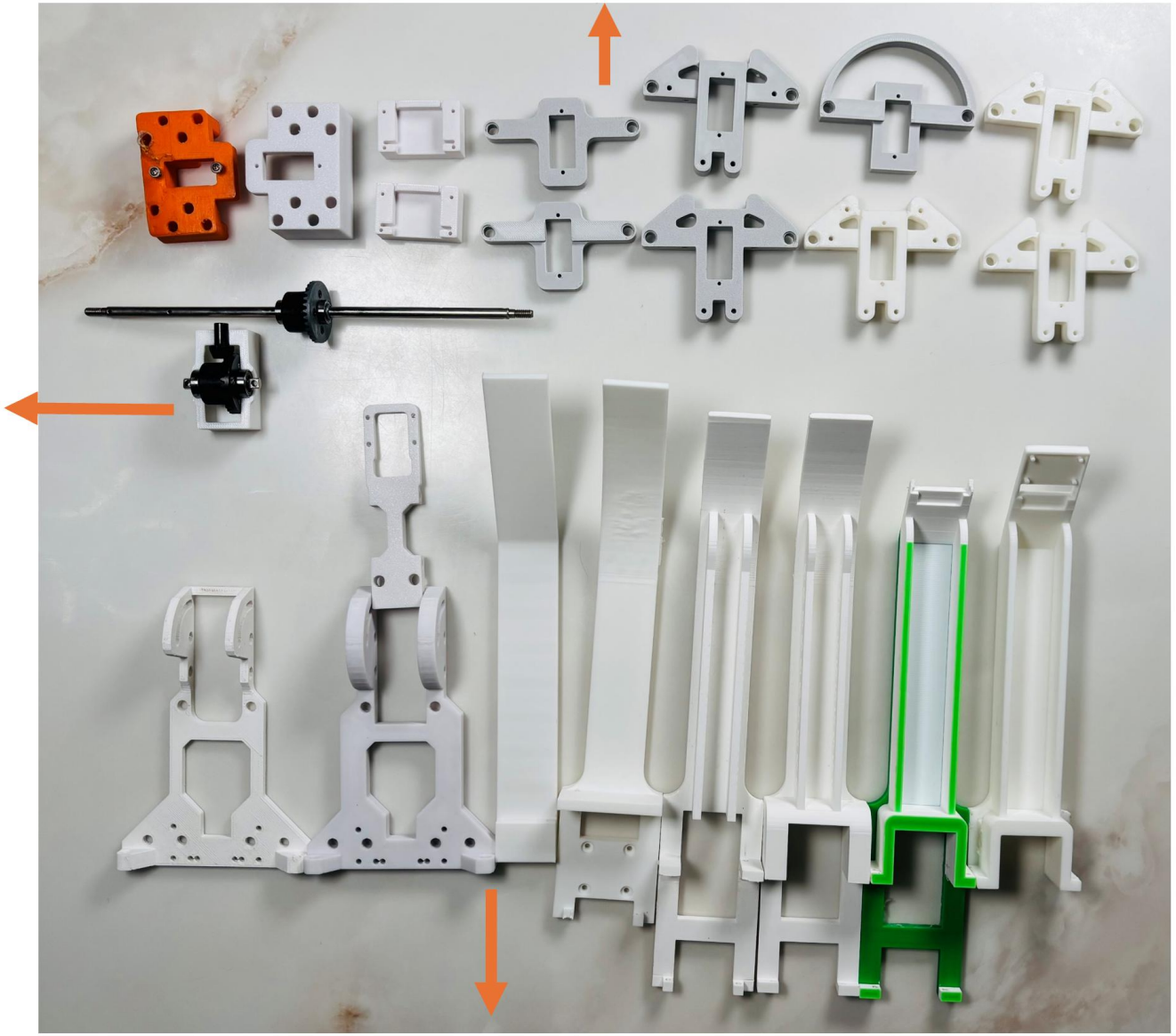


Base Chassis V1-11 From standalone parts to integrated motor mounts and integrated gearbox

Camera & Motor Mount Iterations

Servo Motor Mount- V1-12 from Old to new with changing design to increase stability and avoid touching the wheels

Experiments with Different Gearboxes for multiple Differential Gears



Camera Mount V1-8 from old design to new design with differing curved design, angle, height and base chassis mounting

Power & Sensor Architecture



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Power Architecture



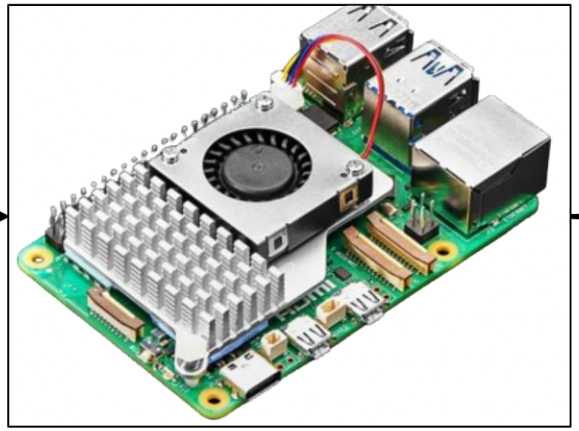
11.1V 1500mAh
16.65 Wh LiPo
Battery

25W/5V Converter
XY-3606

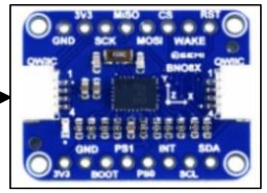


5.2V

Raspberry Pi 5

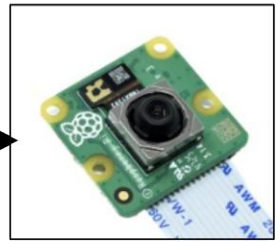


3.3V



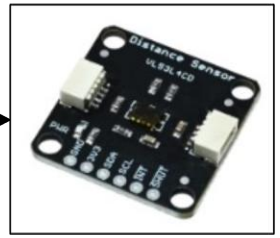
BNO086 IMU

3.3V



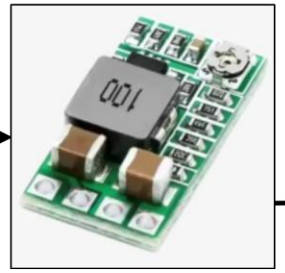
Rpi Camera3 Wide

3.3V



Distance Sensor
4x VL53L4CD.

15 W adj Voltage Converter
MOD-QC2955

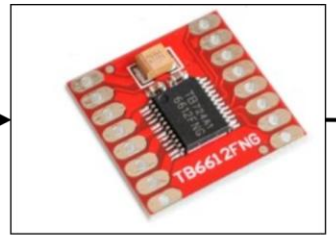


8V



Savox Servo 8V
SV-1232MG

Motor Driver
TB6612FNG

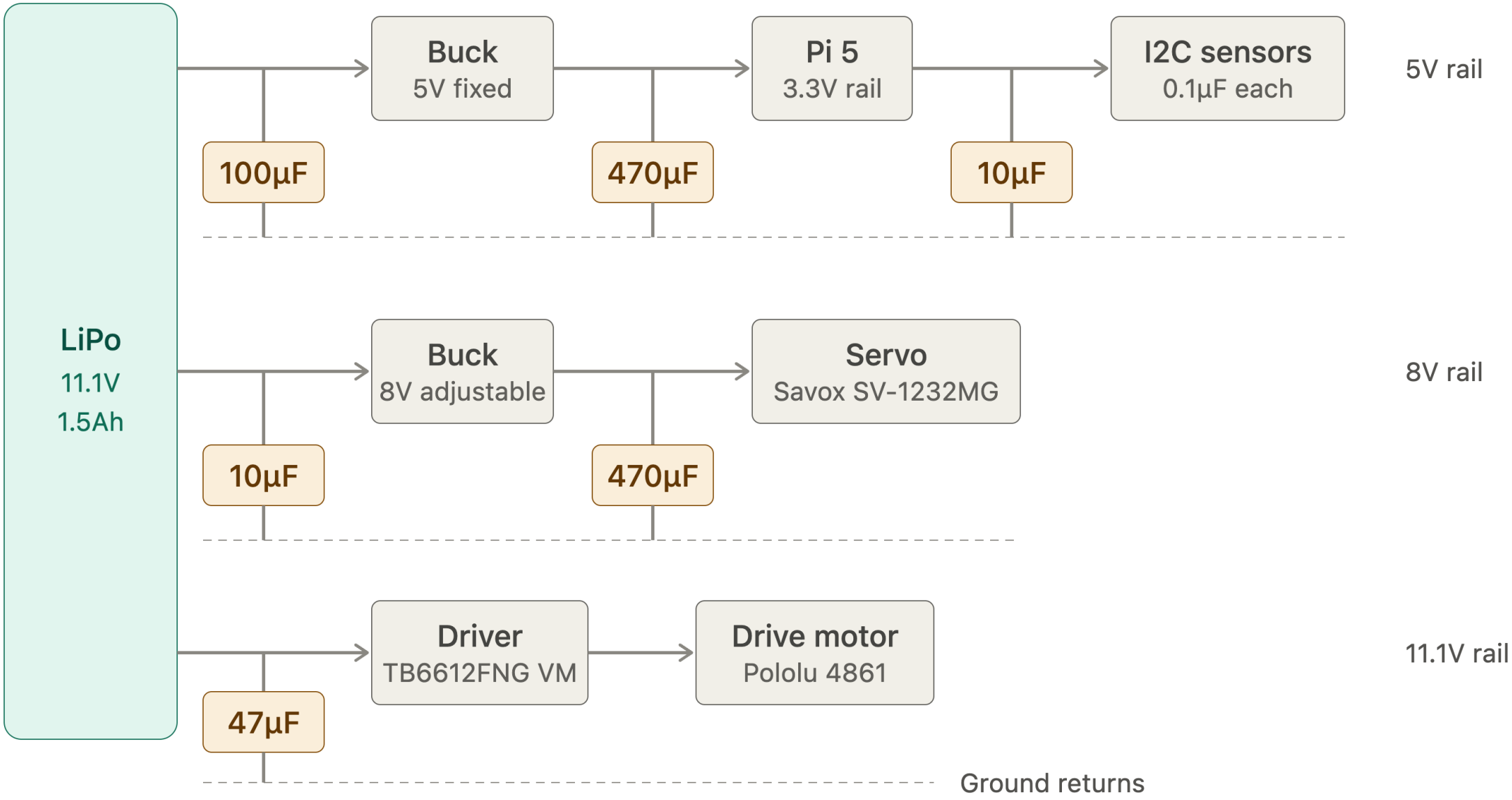


11.1V

Pololu 4861 12V
Drive Motor



Noise Suppression



Power –Current Needs for all components

Devices on 5V power bus	Volts (V)	Idle Current (A)	Typical Current (A)	Peak Current (A)
Raspberry Pi 5	5.0	0.600	0.750	1.6
Camera module 3 wide	5.0	0.250	0.280	0.3
BNO086 IMU	3.3	0.008	0.012	0.015
VL53L4CD ToF sensor (x4)	3.3	0.020	0.080	0.16
Total load on 5V power bus	5.0	0.878	1.122	2.075

Devices on 8V power bus	Volts (V)	Idle Current (A)	Typical Current (A)	Peak Current (A)
Savox Servo- SV1232MG	8.0	0.005	0.200	2.2

Devices on 11V power bus	Volts (V)	Idle (A)	Typical (A)	Peak Current (A)
Motor driver - TB6612FNG (Logic/VCC)	5.0	0.001	0.002	0.003
Motor - Pololu 4861	11.1	0.000	0.400	1.7 (at 11.1V)
Total load on 11V power bus	11.1	0.001	0.402	1.703

* Current values sourced from component datasheets and other sources.

Total Power Calculation

Calculation	Idle Current	Typical Current	Peak Current
5V Bus			
Output power = Voltage × Current	5V × 0.878A = 4.39W	5V × 1.122A = 5.61W	5V × 2.075A = 10.375W
Input power = Output power / Efficiency (95%)	4.39W / 0.95 = 4.62W	5.61W / 0.95 = 5.91W	10.375W / 0.95 = 10.92W
8V Bus			
Output power = Voltage × Current	8V × 0.005A = 0.04W	8V × 0.2A = 1.6W	8V × 2.2A = 17.6W
Input power = Output power / Efficiency (95%)	0.04W / 0.95 = 0.04W	1.6W / 0.95 = 1.68W	17.6W / 0.95 = 18.53W
11V Bus			
Output power = Voltage × Current	11.1V × 0.001A = 0.011W	11.1V × 0.402A = 4.462W	11.1V × 1.703A = 18.903W
Input power = Output power / Efficiency (97.5%)	0.011W / 0.975 = 0.011W	4.462W / 0.975 = 4.577W	18.903W / 0.975 = 19.388W
Battery Total			
Power drawn = 5V input + 8V input + 11V input	4.62 W + 0.04W + 0.011W = **4.67W**	5.91W + 1.68W + 4.577W = **12.17**	10.92W + 18.53W + 19.388W = **48.838W**
Current drawn = Power / Battery voltage (11.1V)	4.67W / 11.1V = **0.421A**	12.17W / 11.1V = **1.096A**	48.838 / 11.1V = **4.40A**

Battery Run Time Calculation

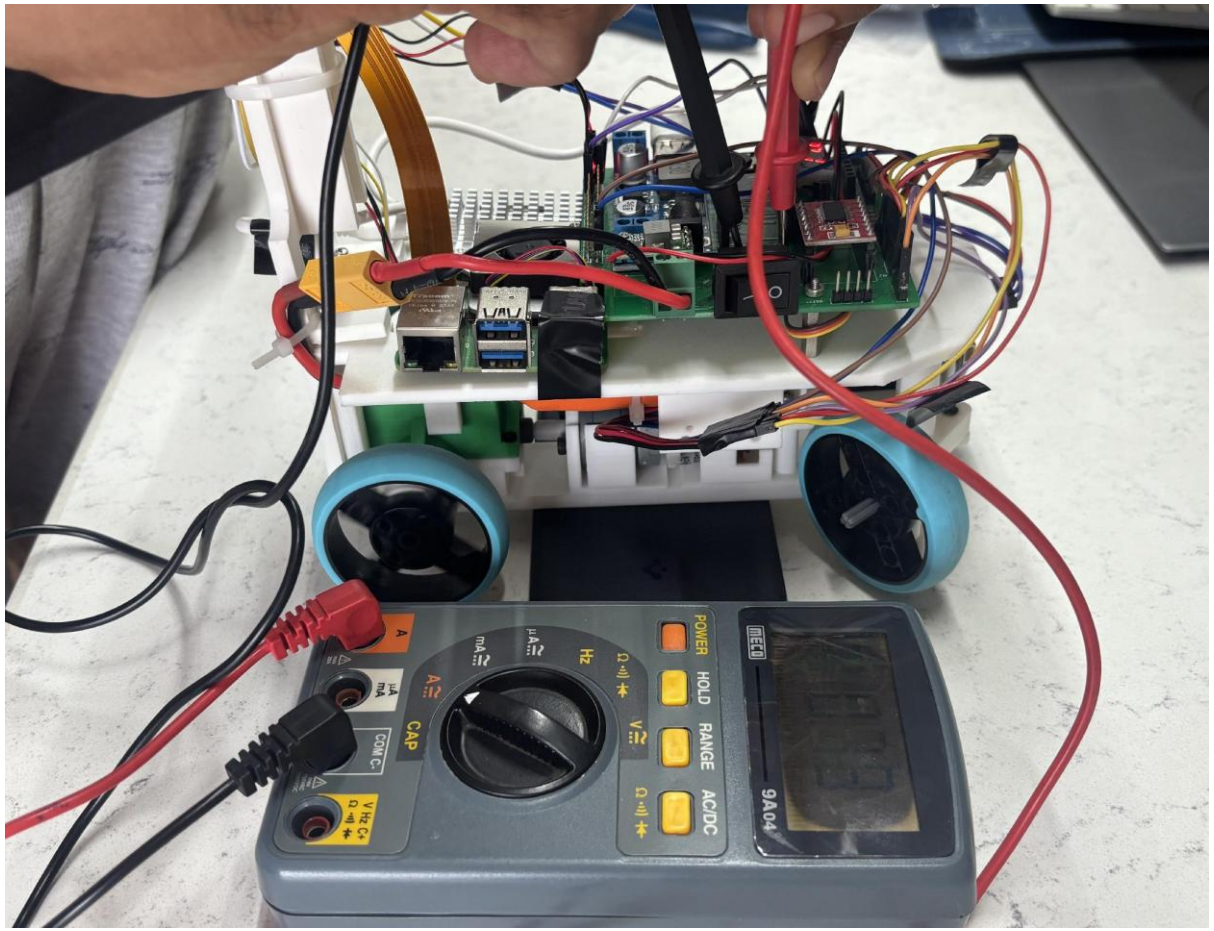
```
Battery specifications
Nominal voltage = 11.1V
Full charge voltage = 12.6V
Capacity              = 1500mAh
Discharge rating = 75C
Max discharge current = capacity * C rating
                                  = 1.5Ah * 75
                                  = 112.5A
Headroom v/s C rating = max discharge current / current drawn from battery
                                  = 112.5A / 1.096A
                                  = 102 x (burst that the battery can handle)

Total energy stored = capacity * nominal voltage
                                = 1.5Ah * 11.1V
                                = 16.65Wh

Estimated runtime = Total energy / Power drawn from battery (typical)
                                = 16.65Wh / 12.17W
                                = 1.37 hours
```

Battery Runtime 1.37 hrs

Verification with Multimeter and Telemetry




```
==========================================================================
SUMMARY  (all values are averages over the 15 s sampling window)
==========================================================================
metric                                  idle_baseline   full_processing
--------------------------------------------------------------------------
Total power (all rails)                       1.930 W          5.706 W
  Onboard power (total - 3V3)                 1.664 W          4.834 W
  3V3_SYS rail power                          0.266 W          0.872 W
  3V3_SYS rail current                       0.0800 A         0.2626 A
  3V3_SYS rail voltage                       3.3203 V         3.3210 V
--------------------------------------------------------------------------
CPU usage avg                                 0.1 %           46.6 %
  CPU usage min / max                   0.0 / 0.9 %     14.1 / 57.7 %
CPU temp avg                                 39.3 C           46.3 C
  CPU temp min / max                   38.4 / 41.1 C   41.7 / 47.7 C
ARM clock avg                              1596 MHz         2361 MHz
  ARM clock min / max                1500 / 2400 MHz  1800 / 2400 MHz
RAM used avg                                513 MB           747 MB
  RAM used min / max                 501 / 531 MB     709 / 756 MB
throttled flags                               0x0              0x0
--------------------------------------------------------------------------

Delta (full_processing - idle_baseline):
  total power    +3.777 W
  3V3 rail       +0.606 W
  CPU usage      +46.5 %
  CPU temp       +6.9 C
```

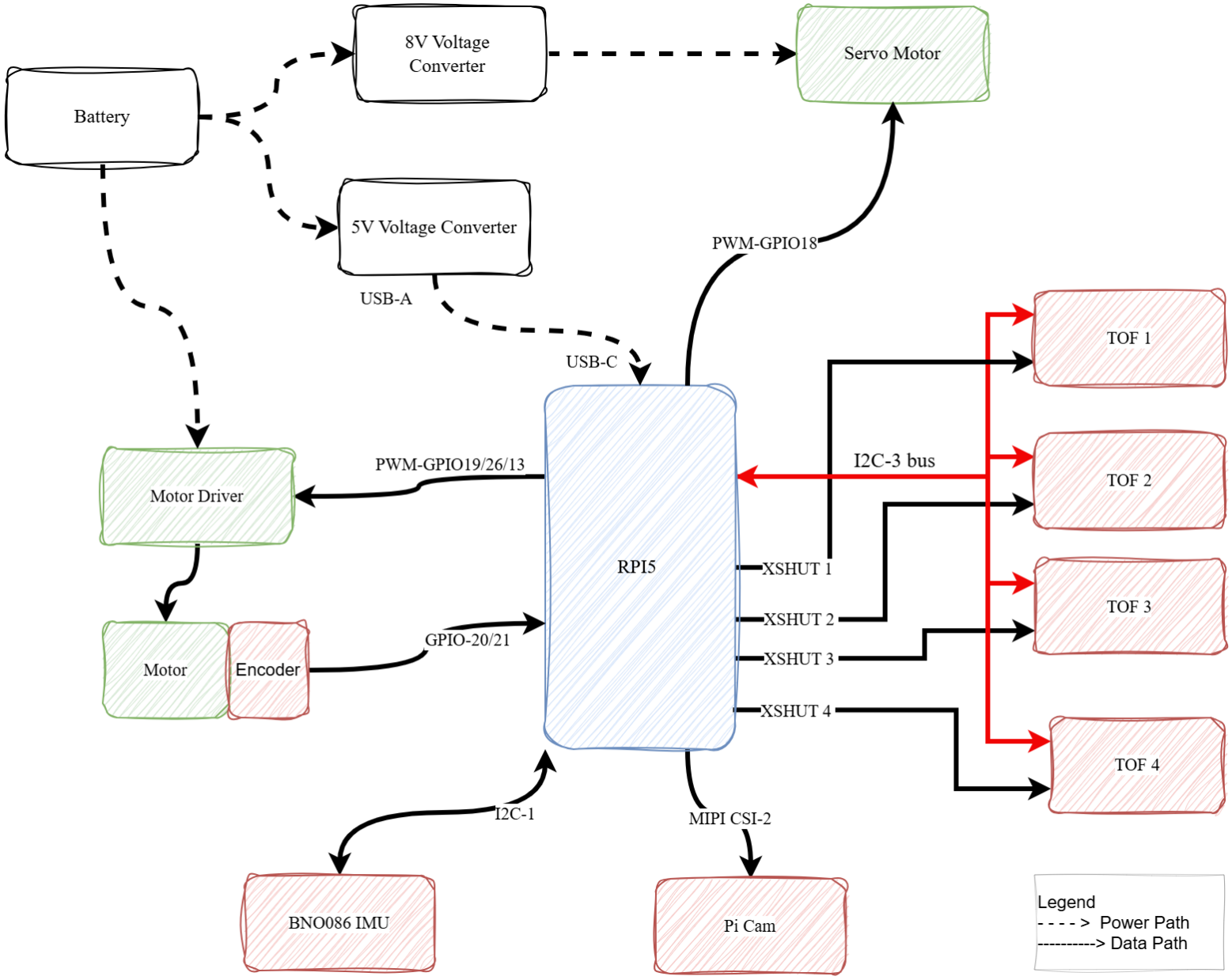
Sum of current(typical) for Rpi & sensors** = 1.062A

Total Power consumed by Rpi = 1.062A * 5V = 5.31W

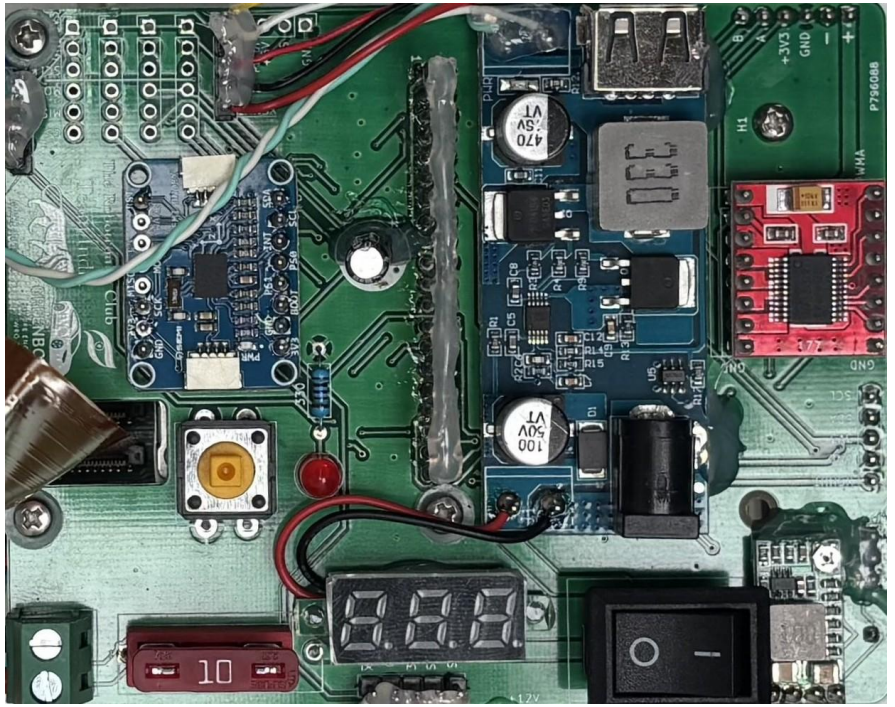
** Excludes current for Servo as it is drawn directly from converter

Measured Power = Calculated Power

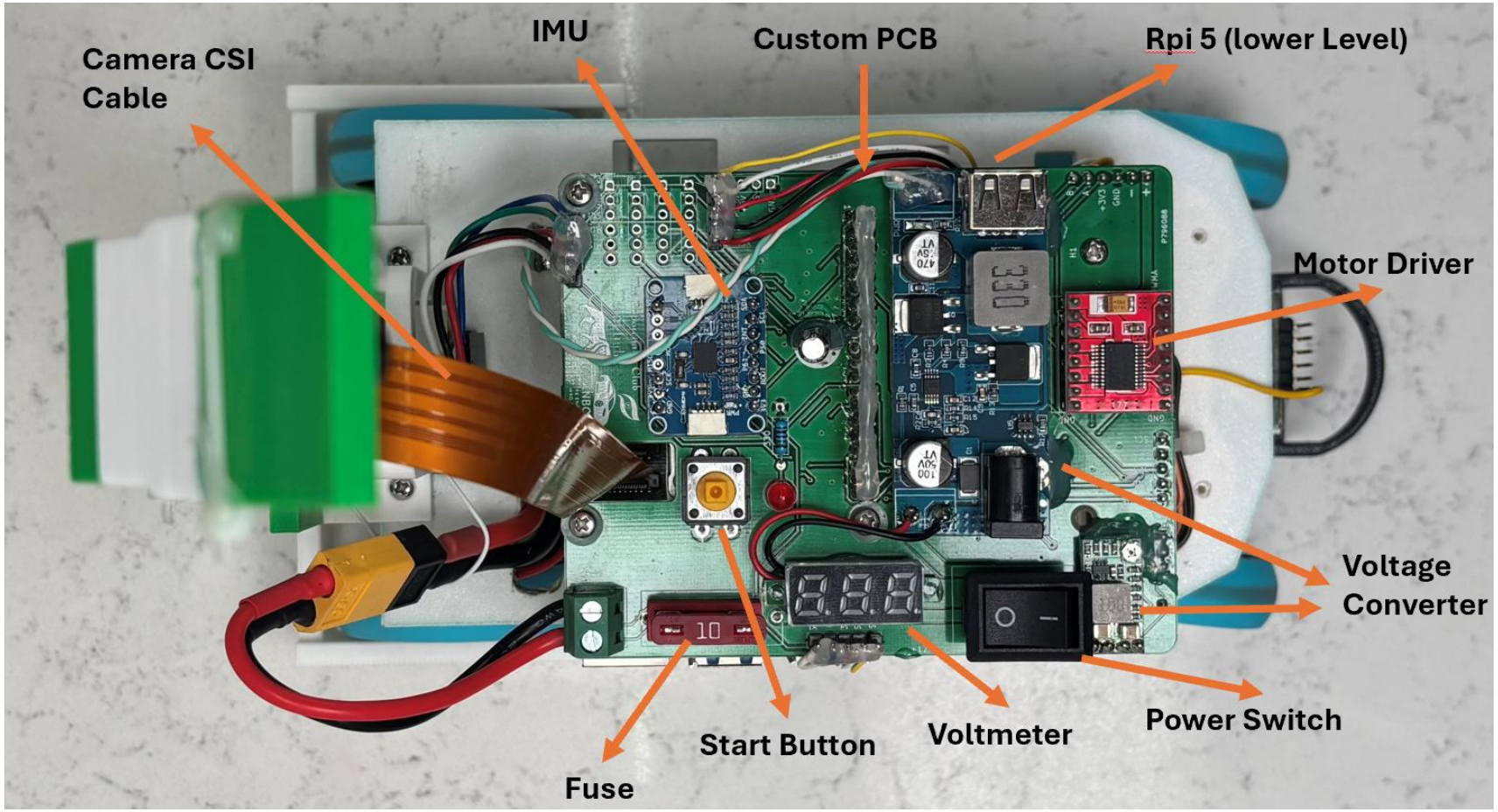
Simplified Wiring Diagram



Custom PCB



Electrical Safety Features



Voltmeter

Fuse

Power Rail Isolation

Distance Sensor Selection

3.7.3 Distance Sensor

Specification Comparison

Specification	VL53L4CD	VL53L1X	VL53L8CX
Zone Architecture	Single Zone	Single Zone *(Programmable ROI)*	**Multi-Zone (4x4 or 8x8 grid)**
Maximum Range	Up to 1.3 meters (1300 mm)	**Up to 4.0 meters (4000 mm)**	**Up to 4.0 meters (4000 mm)**
Minimum Range	**Down to 1 mm**	~40 mm	~20 mm
Field of View (FoV)	18° Diagonal	15° to 27° (Adjustable via ROI)	**65° Diagonal** (Wide angle)
Max Sampling Rate	**Up to 100 Hz**	Up to 50 Hz	Up to 60 Hz
Ambient Light Immunity	Moderate	Standard	**Excellent** (Uses histogram processing)
Primary Focus	Ultra-short range precision & speed	Mid-to-Long range point sensing	Scene mapping, multi-target tracking

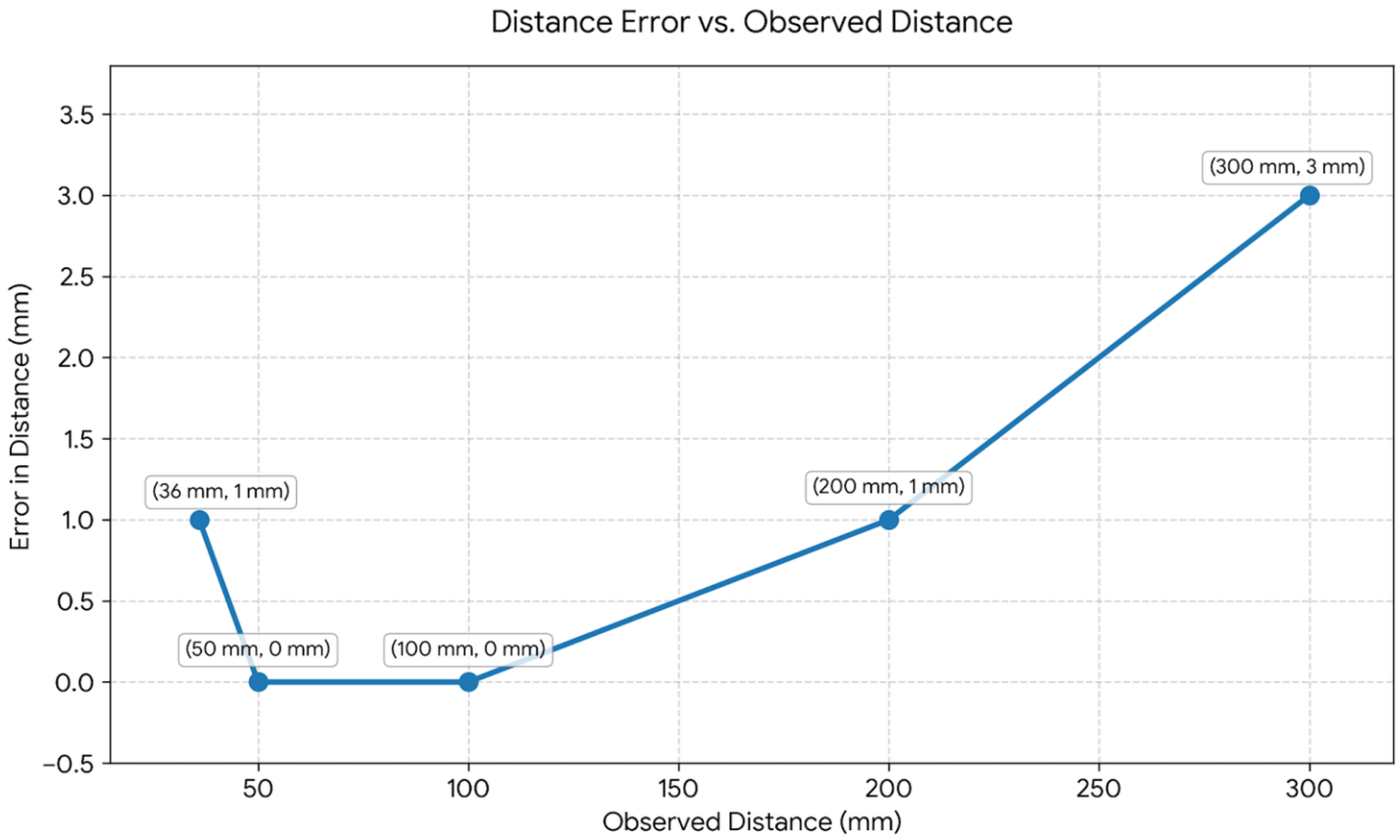
High Precision

Faster Reaction Time

Distance Sensor Selection

Experiment

Experiment #	ToF measured distance (mm)	Observed distance practically (mm)	Δ distance (mm)
1	303	300	3
2	201	200	1
3	100	100	0
4	50	50	0
5	35	36	1
6	12	5	7

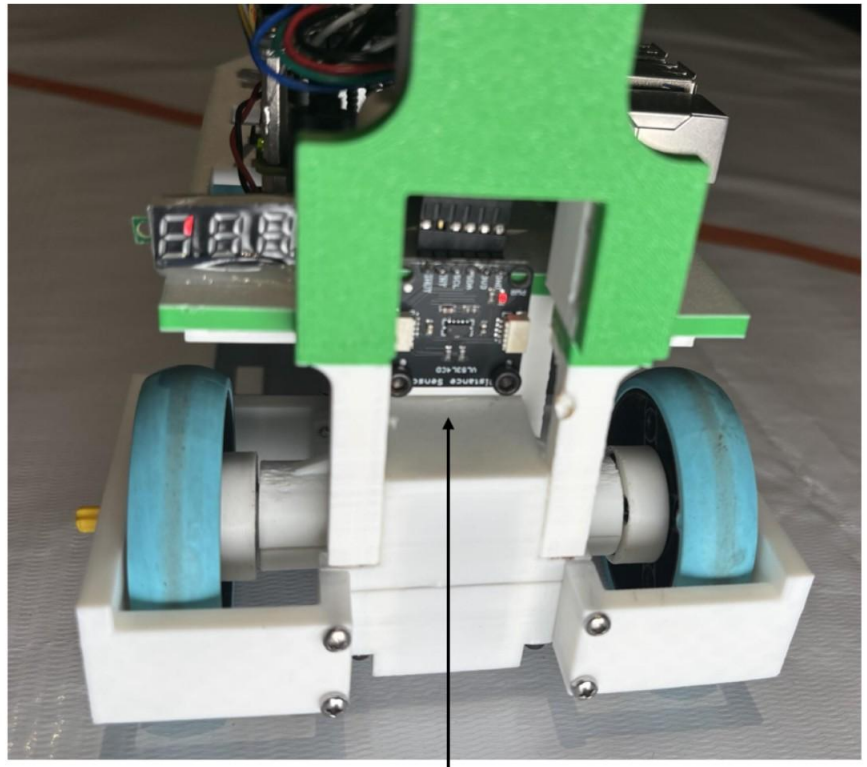


Measured vs Actual distance

Blind spot at very short range

Distance Sensor Selection

Placement

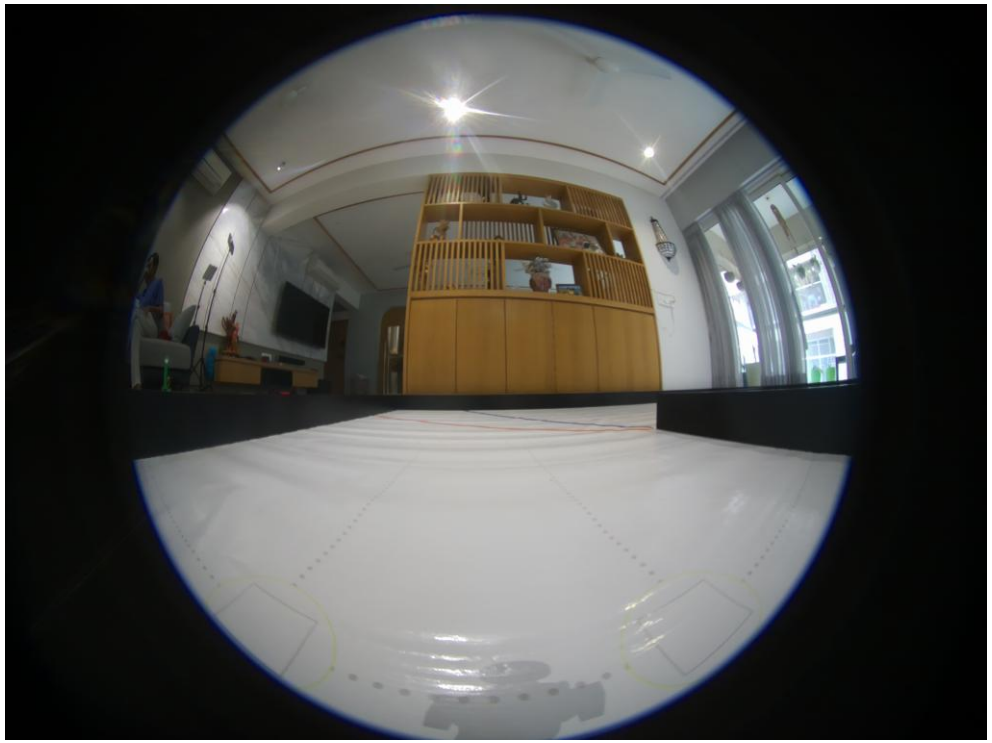


Recessed TOF Sensor

Avoid the blind spot

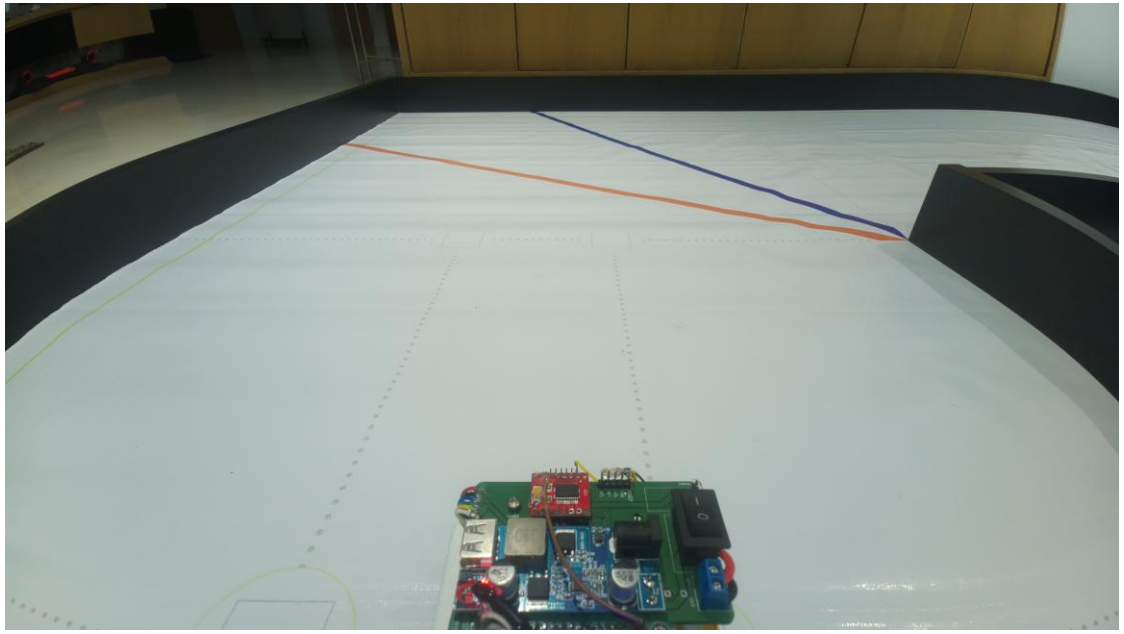
Mount recessed

Camera Selection



HQ + 180° fisheye

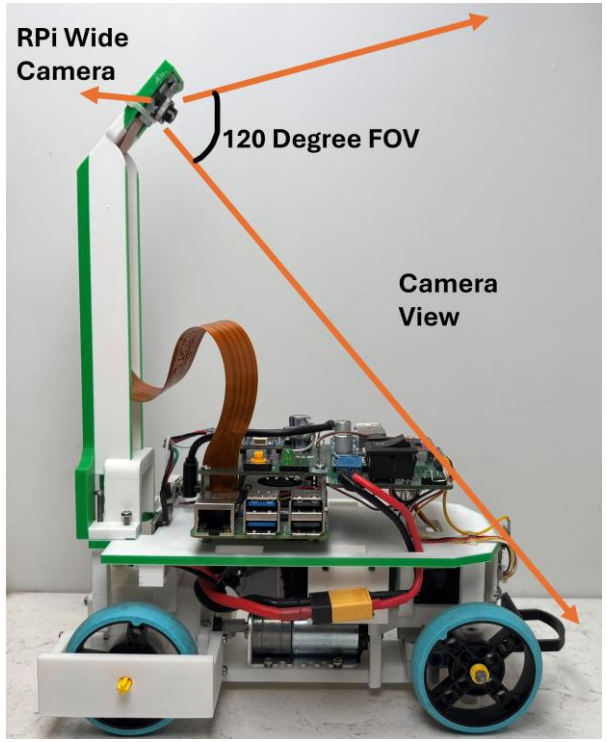
Distorted edges

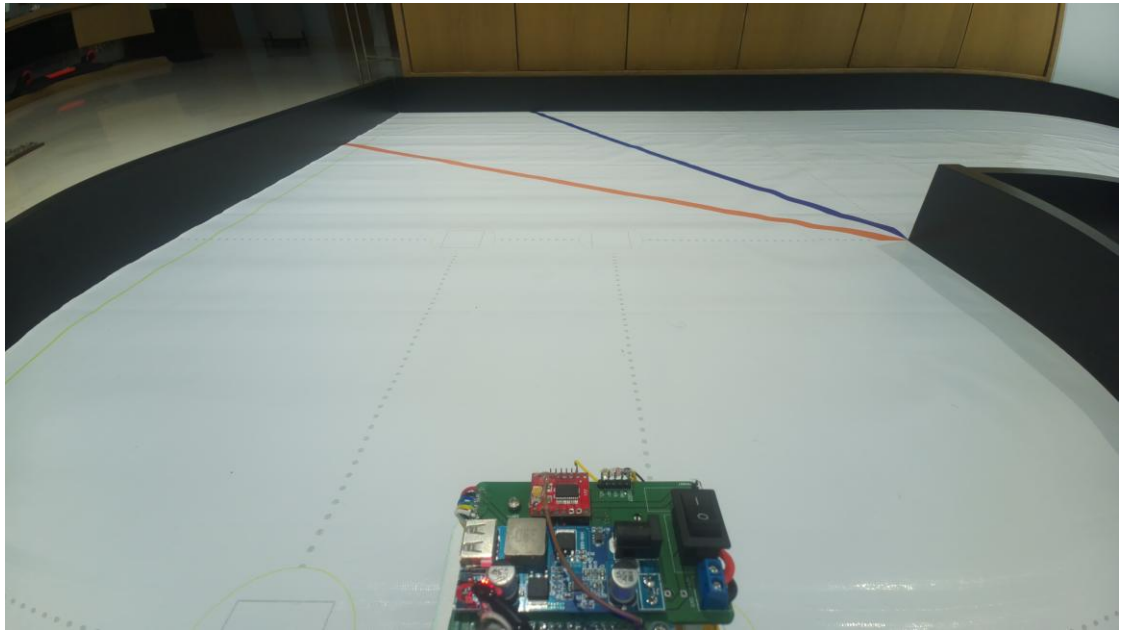


Camera Module 3 Wide

120° FOV, no edge distortion

Camera Selection



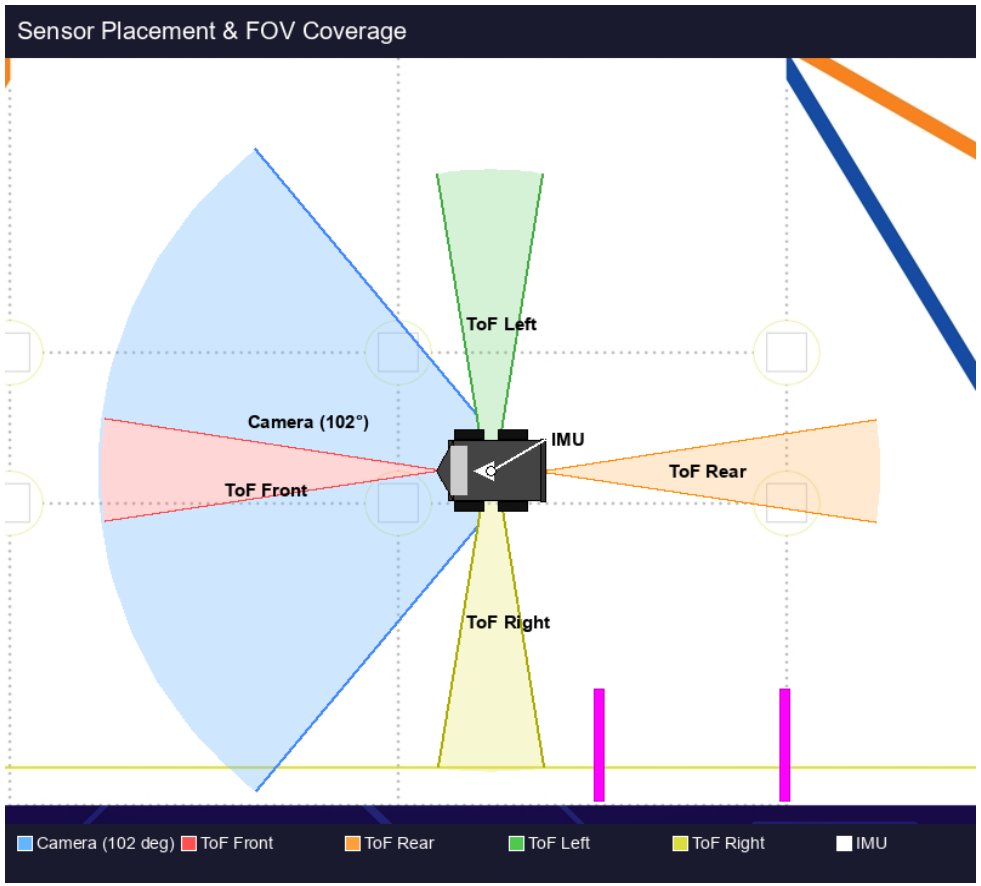


Placement guided by FOV

Angle keeps front edge in view

Camera Module 3 Wide

120° FOV, no edge distortion



Sensor Placement

Camera
26 cm high
102° H × 67° V

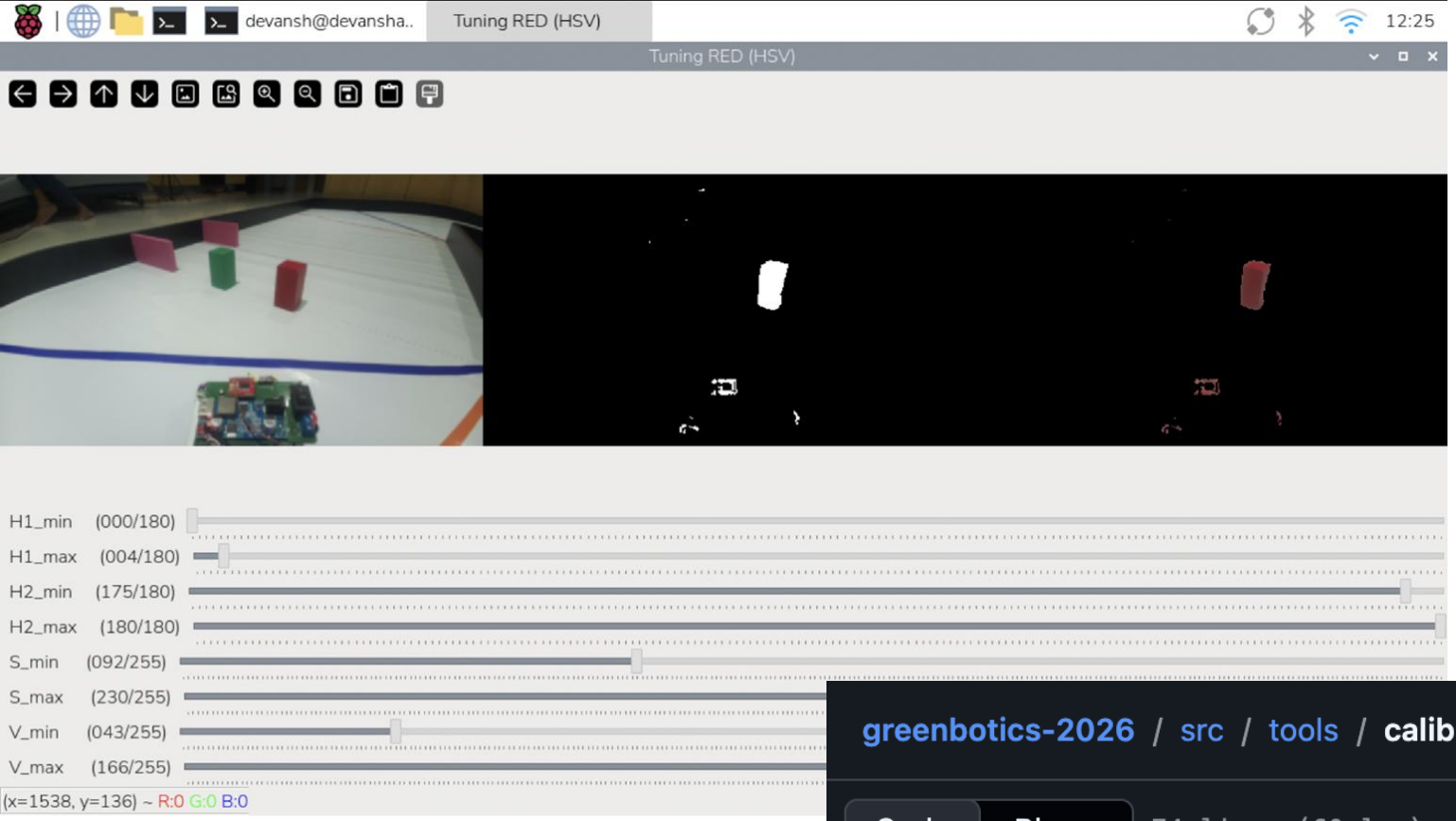
FOV

IMU
centre of chassis · 11.6 cm

ToF Left · 5.5 cm
90° left, recessed (faces viewer)

ToF Rear · 6.7 cm

ToF Front · 5.5 cm



HSV calibration

IMU calibration

greenbotics-2026 / src / tools / calibrate_bno.py

Code | Blame | 74 lines (60 loc) · 2.87 KB

```
17  def run_calibration():
37      print("\n--- Starting Calibration Mode ---")
38      sensor.begin_calibration()
39      print("Self-calibration triggered.")
40
41      print("\nCalibration instructions:")
42      print("1. GYROSCOPE: Place the robot on a flat, vibration-free surface and keep it
43      print("2. ACCELEROMETER: Slowly rotate the robot so that it rests on each of its 6
44      print("3. MAGNETOMETER: Move/rotate the robot in a figure-8 motion in the air.")
45      print("\nCalibration status levels: 0=Unreliable, 1=Low, 2=Medium, 3=High")
46      print("Press Ctrl+C at any time to save the calibration data and exit.")
```

Software Architecture & Obstacle Strategy



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State Machine Diagram

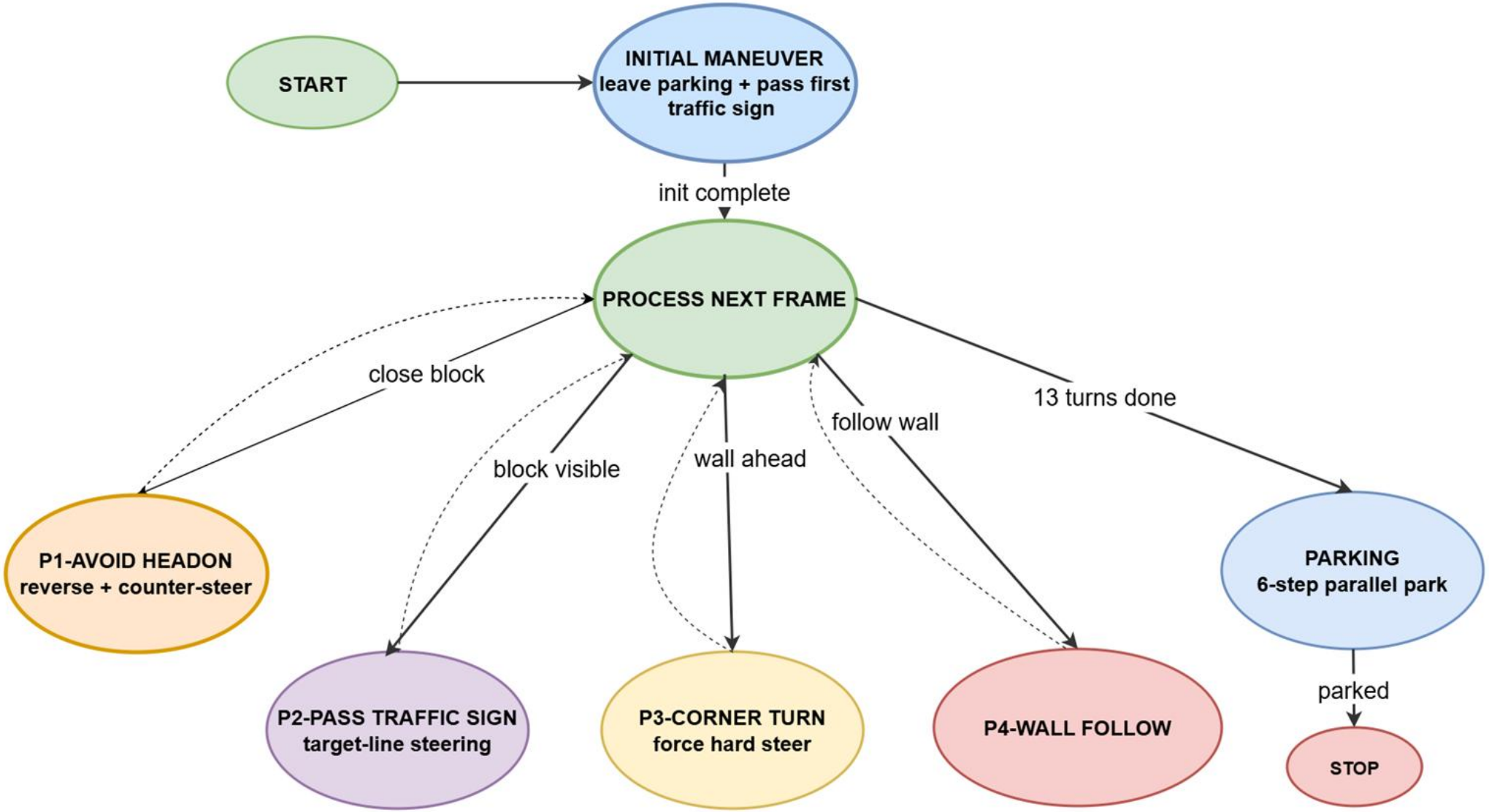
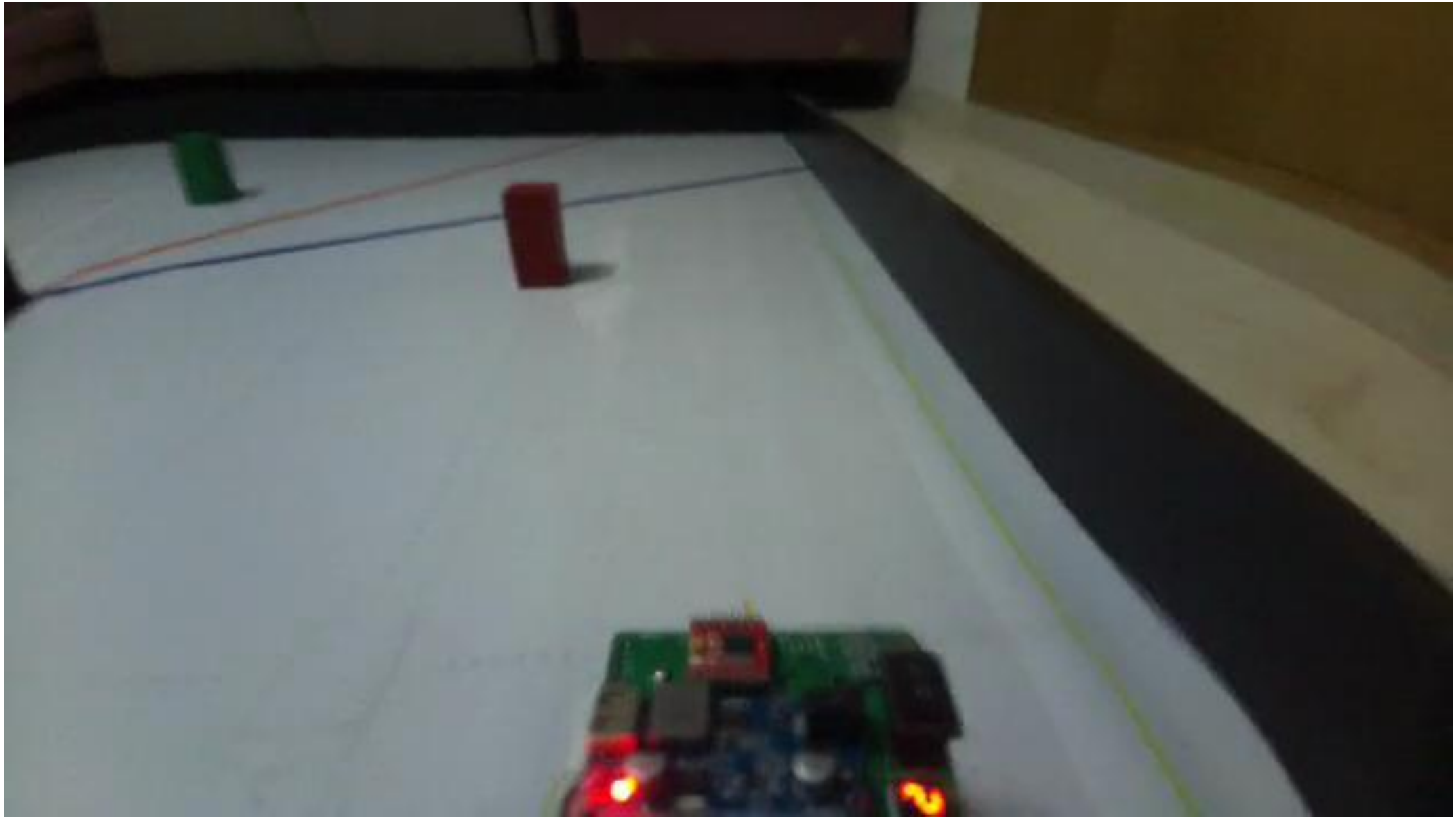


Image Processing



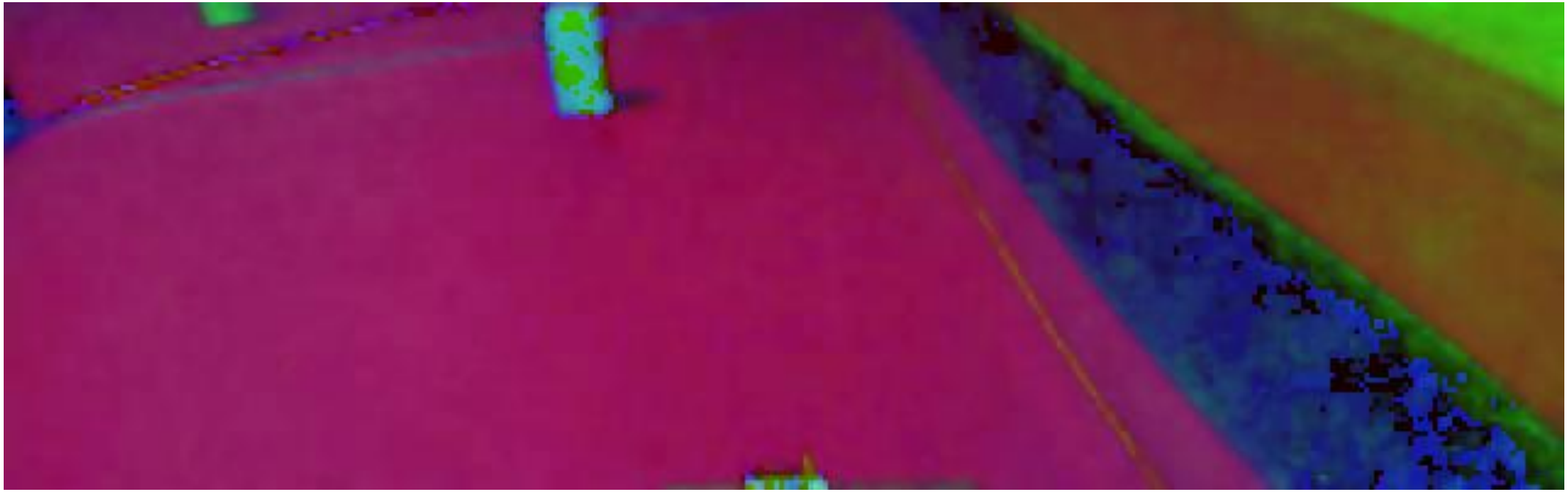
Raw

Image Processing



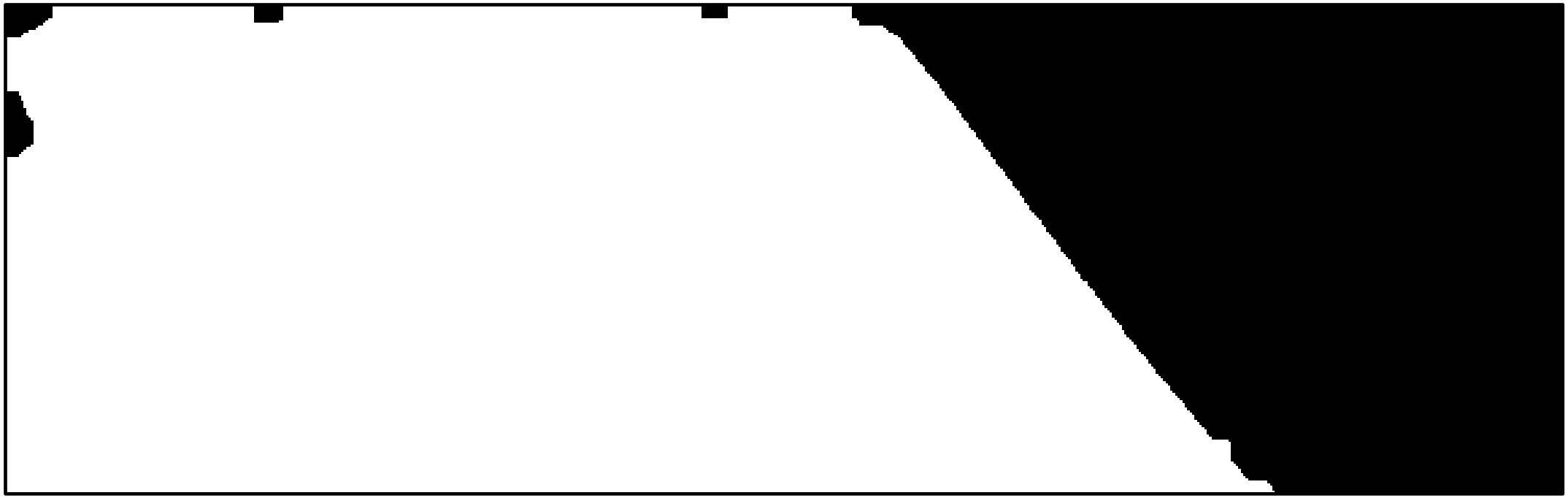
Crop

Image Processing



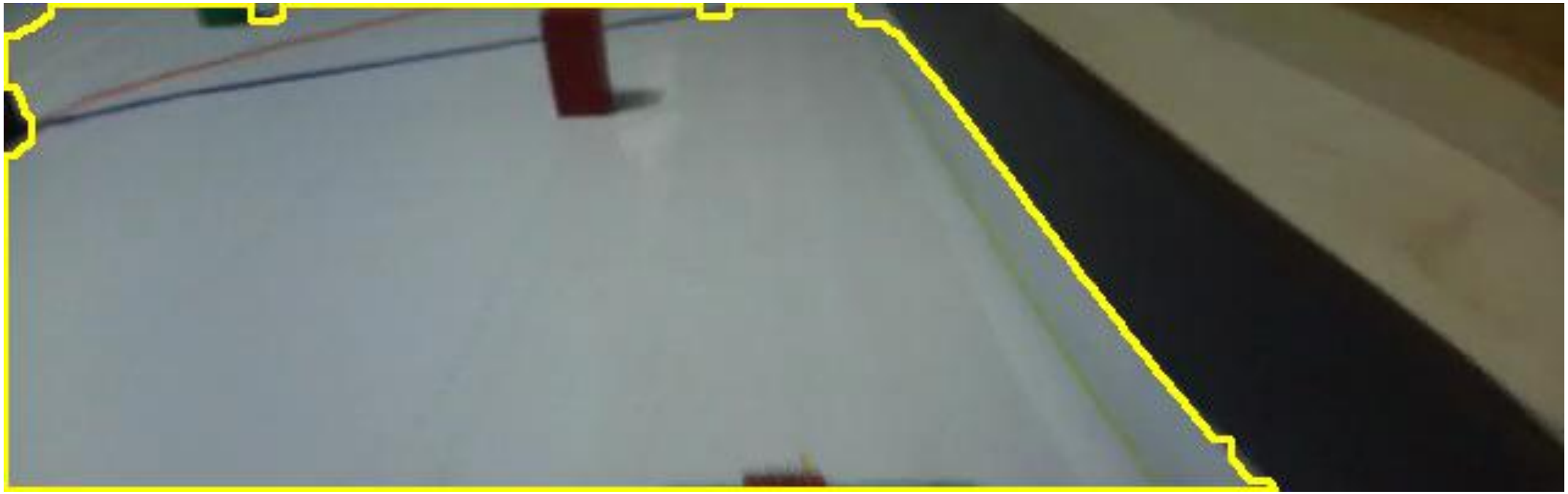
HSV

Image Processing



White Mask

Image Processing



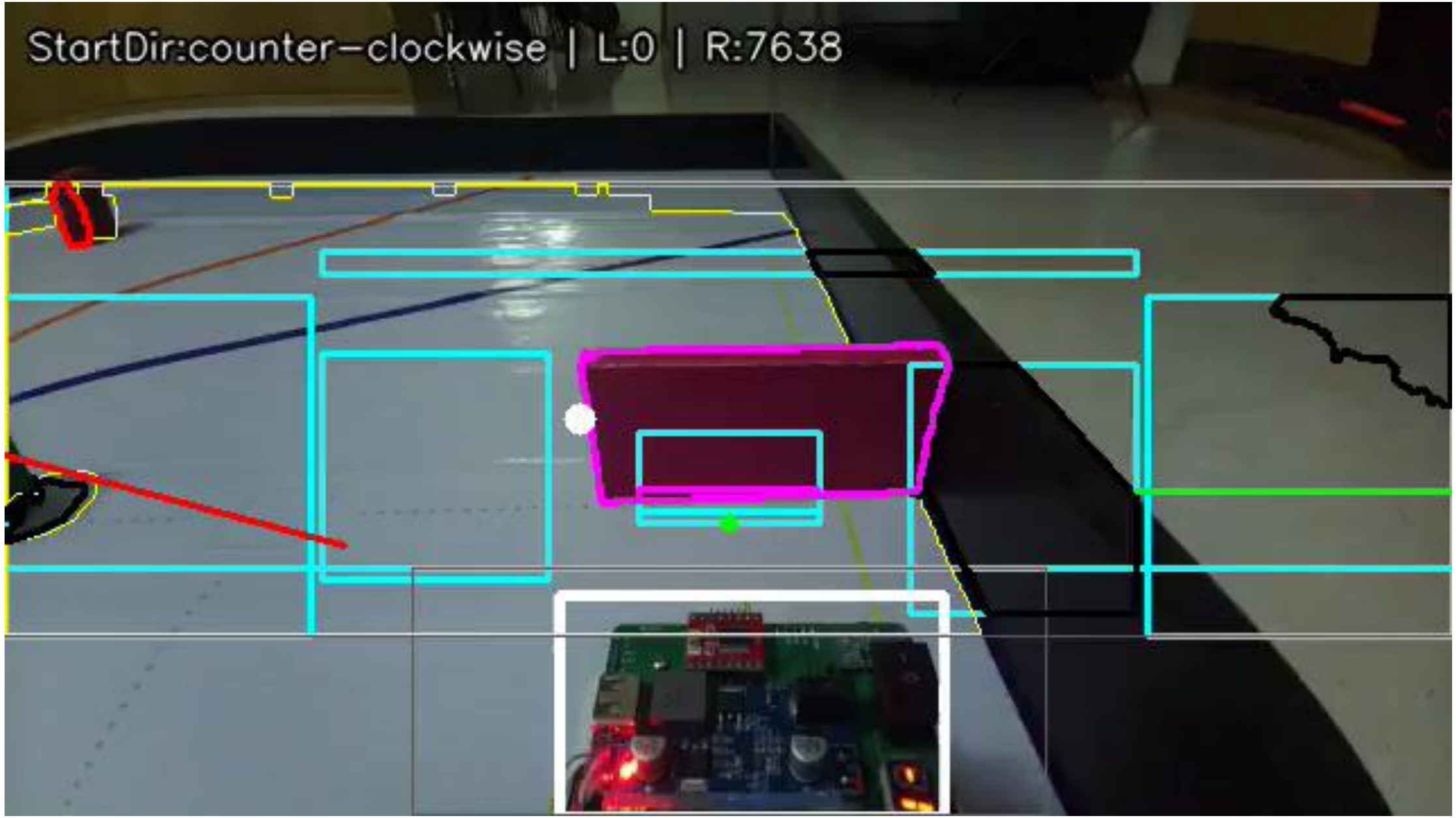
Arena Outline

Image Processing



Color Mask

Robot FPV

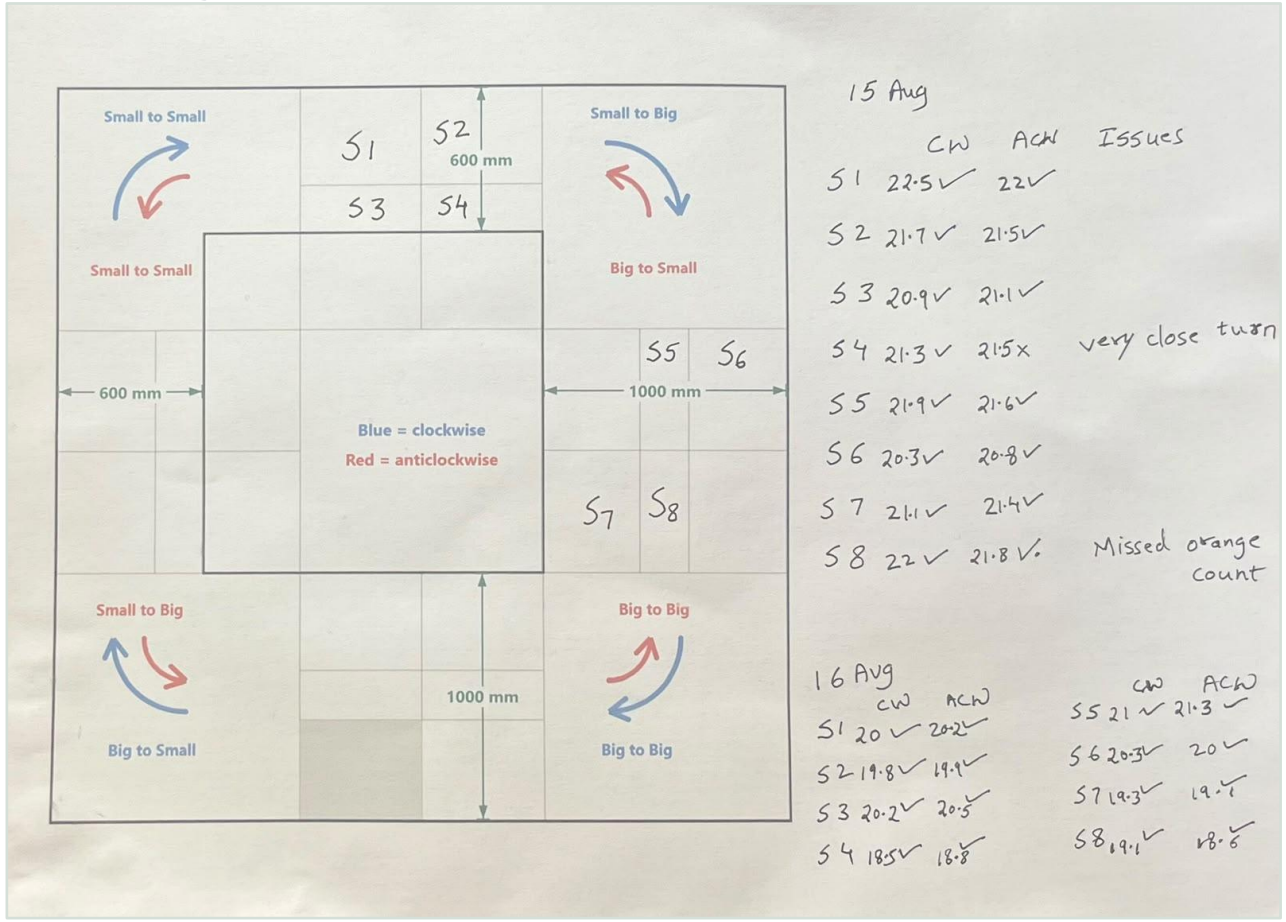


Open Challenge Testing

32 runs

8 start sections · avg 20.6 s · 15–16 Aug

Every turn combination



Run log, S1–S8, clockwise and anti-clockwise

1. **Small to small**
2. **Small to big**
3. **Big to small**
4. **Big to big**

600 mm and 1000 mm tracks · both directions

15 Aug — **Round 1 · 16 runs · avg 21.5 s**
Close corners and a missed orange count → tuned

16 Aug — **Round 2 · 16 runs · avg 19.8 s**
Full set re-run → all passed

Round 1 close corners and missed orange count → tuned → full re-run, all passed

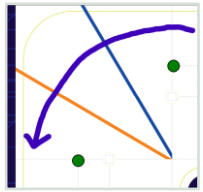
Phase 1 › Phase 2 › Phase 3

Obstacle Challenge Testing

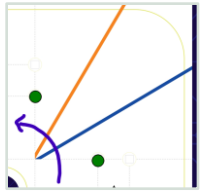
24 runs

6 layouts · avg 47 s · 28 Jul – 9 Aug

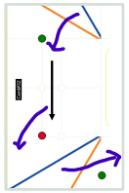
Hardest layouts first



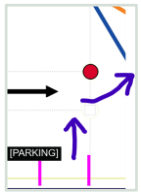
1 **Tight outer**



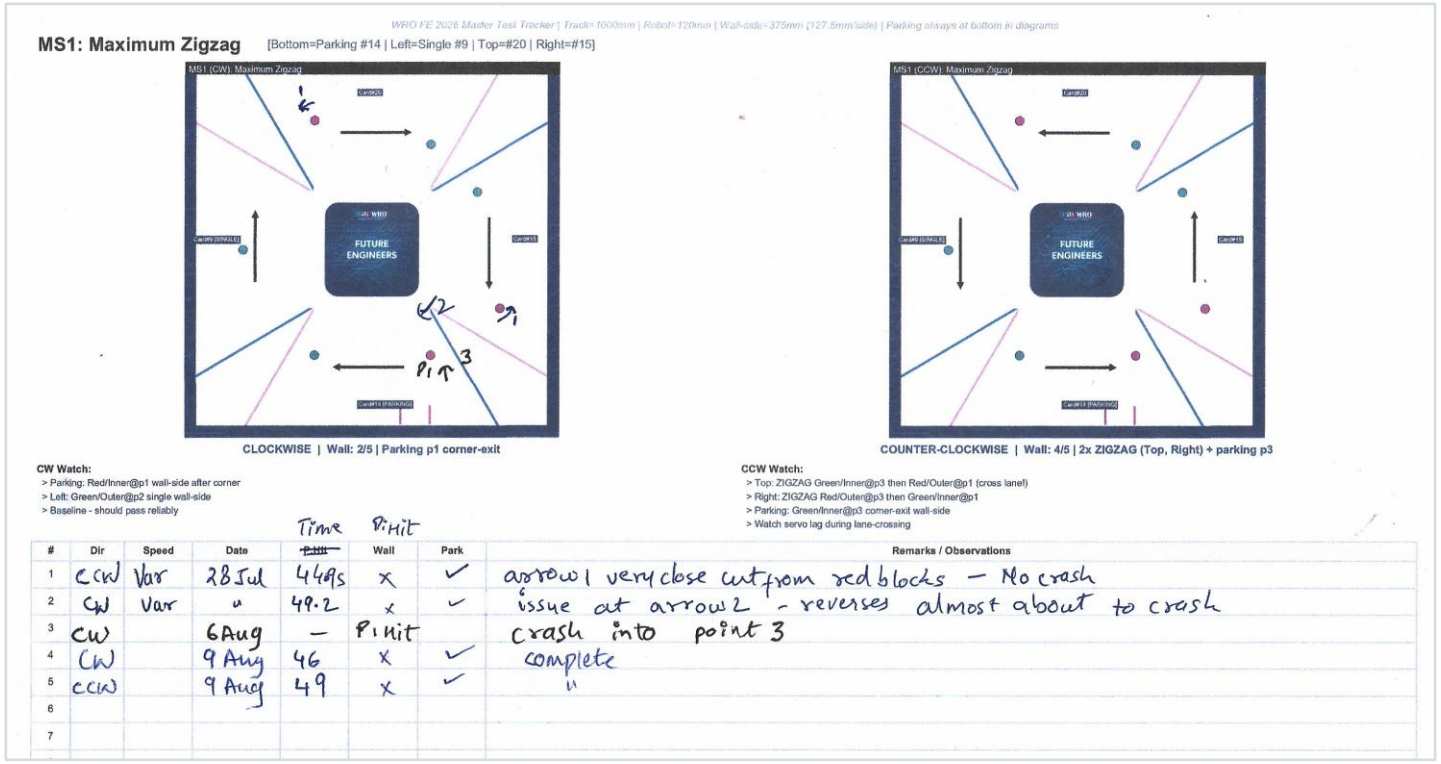
2 **Tight inner**



3 **Max zigzag**



4 **Parking exit blocked**



MS1: Maximum Zigzag [Bottom=Parking #14 | Left=Single #9 | Top=#20 | Right=#15]

CLOCKWISE | Wall: 2/5 | Parking p1 corner-exit

COUNTER-CLOCKWISE | Wall: 4/5 | 2x ZIGZAG (Top, Right) + parking p3

CW Watch:
- > Parking: Red/Inner@p1 wall-side after corner
- > Left: Green/Outer@p2 single wall-side
- > Baseline - should pass reliably

CCW Watch:
- > Top: ZIGZAG Green/Inner@p3 then Red/Outer@p1 (cross lane!)
- > Right: ZIGZAG Red/Outer@p3 then Green/Inner@p1
- > Parking: Green/Inner@p3 corner-exit wall-side
- > Watch servo lag during lane-crossing

#	Dir	Speed	Date	Time ~~Pi Hit~~	Pi Hit Wall	Park	Remarks / Observations
1	CCW	Var	28 Jul	44.9s	x	✓	arrow 1 very close cut from red blocks – No crash
2	CW	Var	"	49.2	x	✓	issue at arrow 2 - reverses almost about to crash
3	CW		6 Aug	–	Pi Hit		crash into point 3
4	CW		9 Aug	46	x	✓	complete
5	CCW		9 Aug	49	x	✓	"
6							
7							

Run sheet, MS1

Wall contacts, near crashes in reverse and one full crash - each troubleshot and fixed.

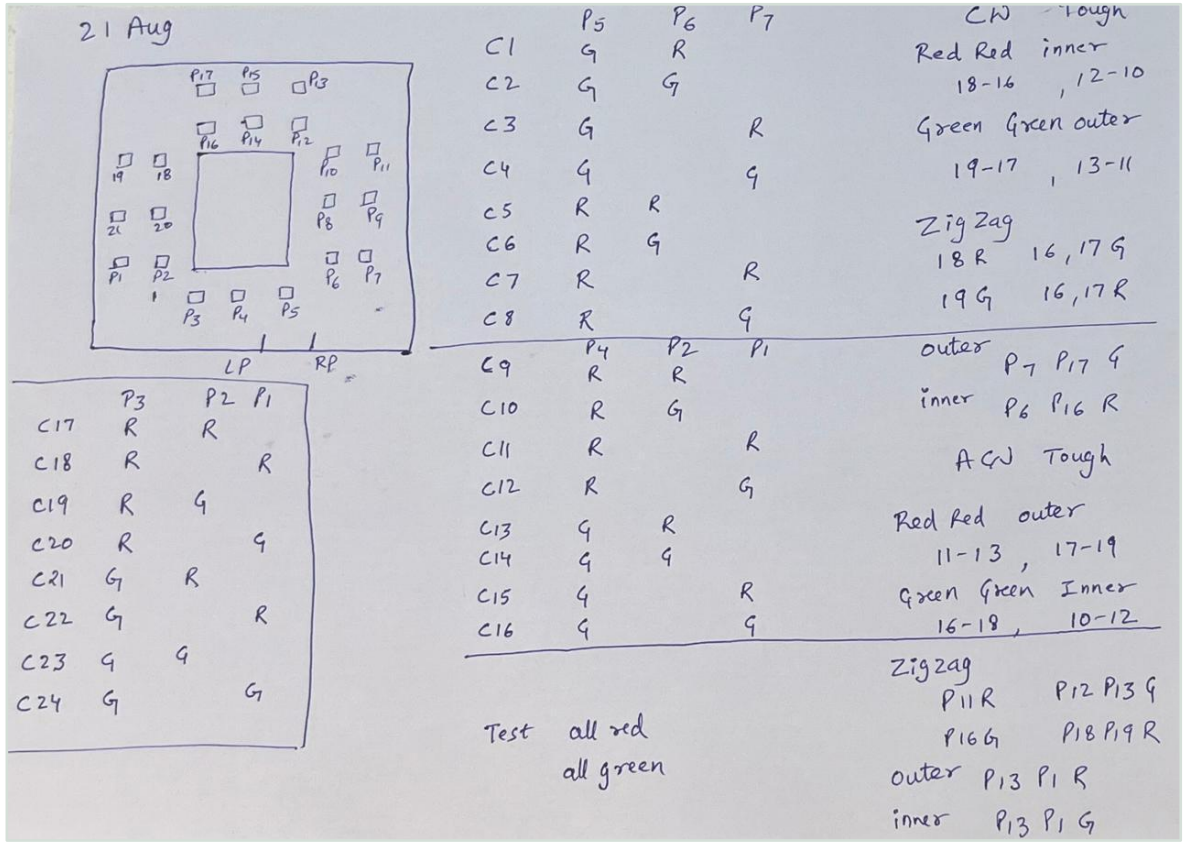
Phase 1 › **Phase 2** › Phase 3

Obstacle Challenge Testing

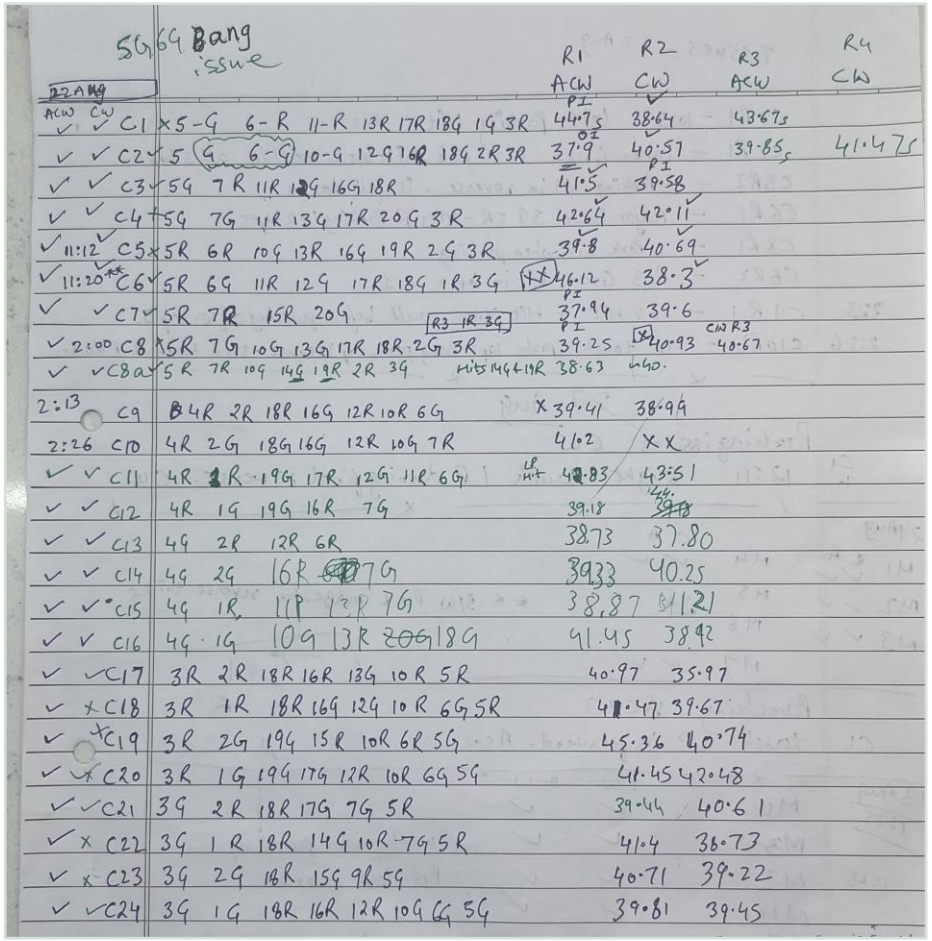
105 runs

55 + 50 re-runs · 24 layouts · avg 40.2 s · 21–22 Aug

Every parking exit & Difficult Placement



24 combinations



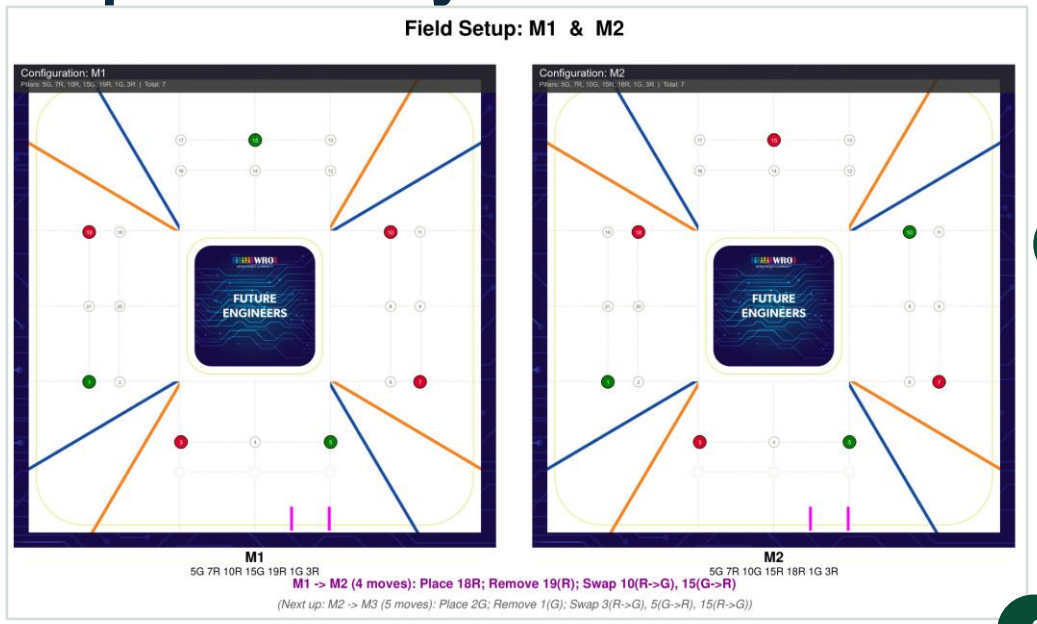
Run log

Defects found in parking, inner turns, passing between parking and block→ fixed → re-run in full.

Phase 1 › Phase 2 › **Phase 3**

Obstacle Challenge Testing

7 optimized layouts

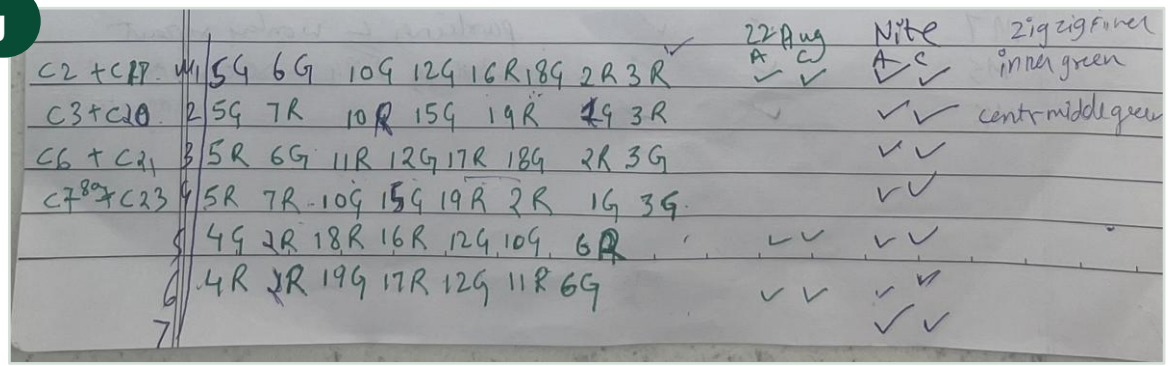


Run-book: 4 min, 4 runs

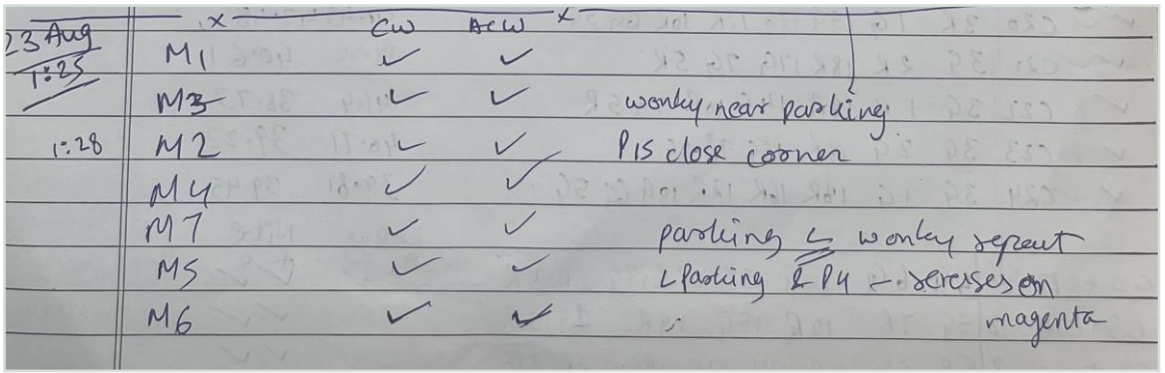
Summary 169 runs

in 4 weeks · avg 42 s per run

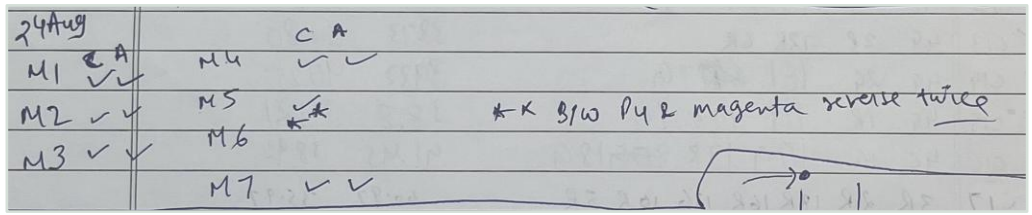
22 Aug



23 Aug



24 Aug



Optimized 7 layouts, run multiple times over 3 days with minor adjustments- all passed

Systems Thinking & Engineering Decisions

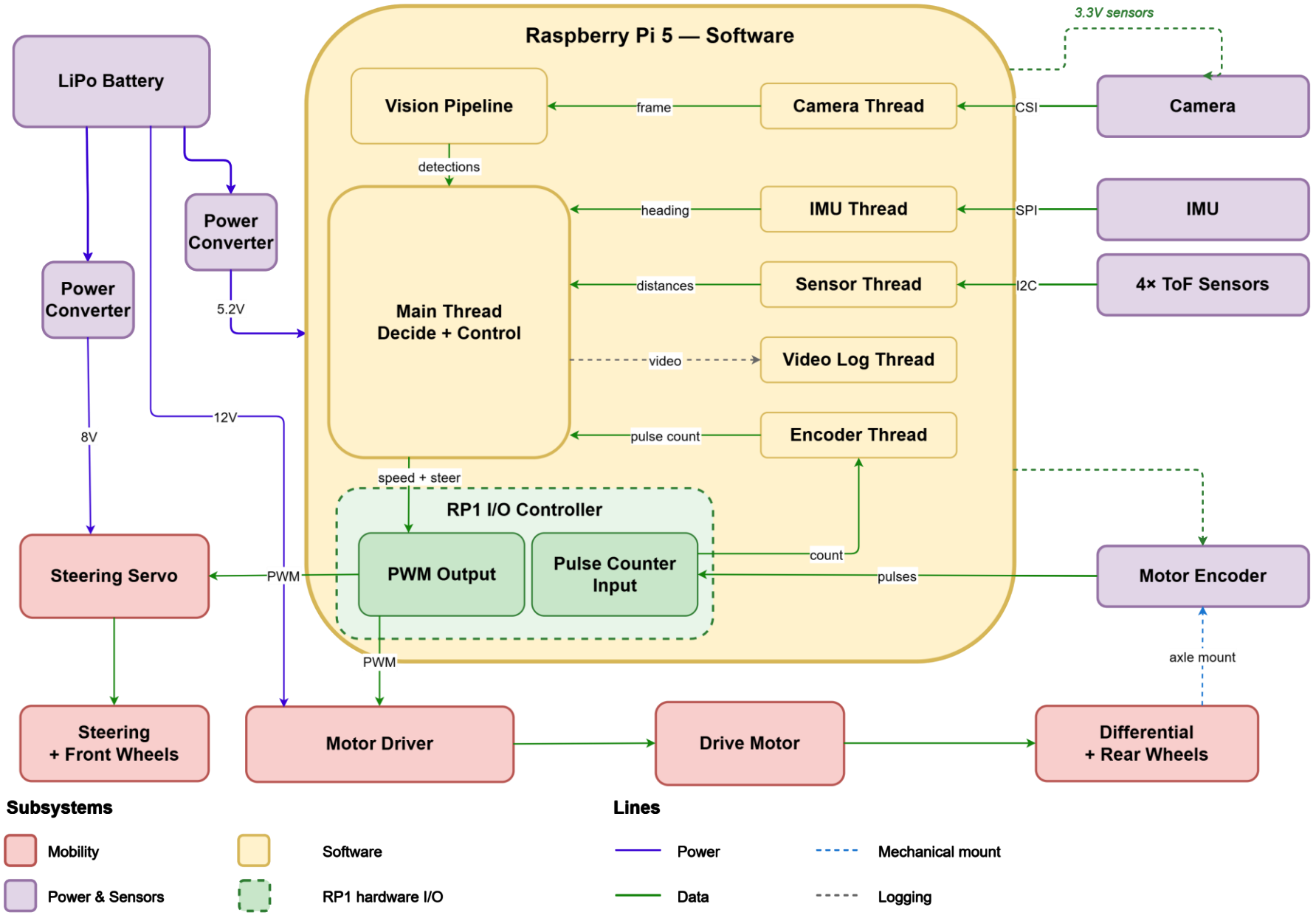


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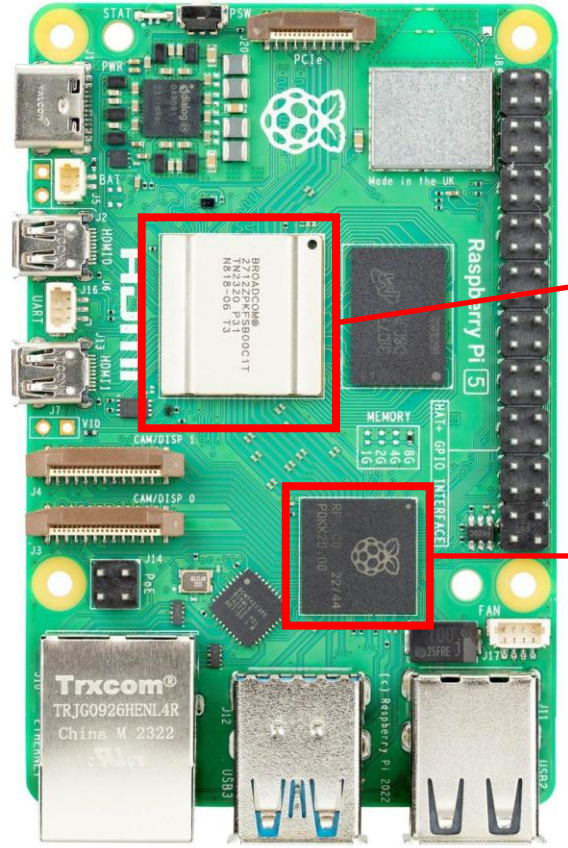
WRO 2026

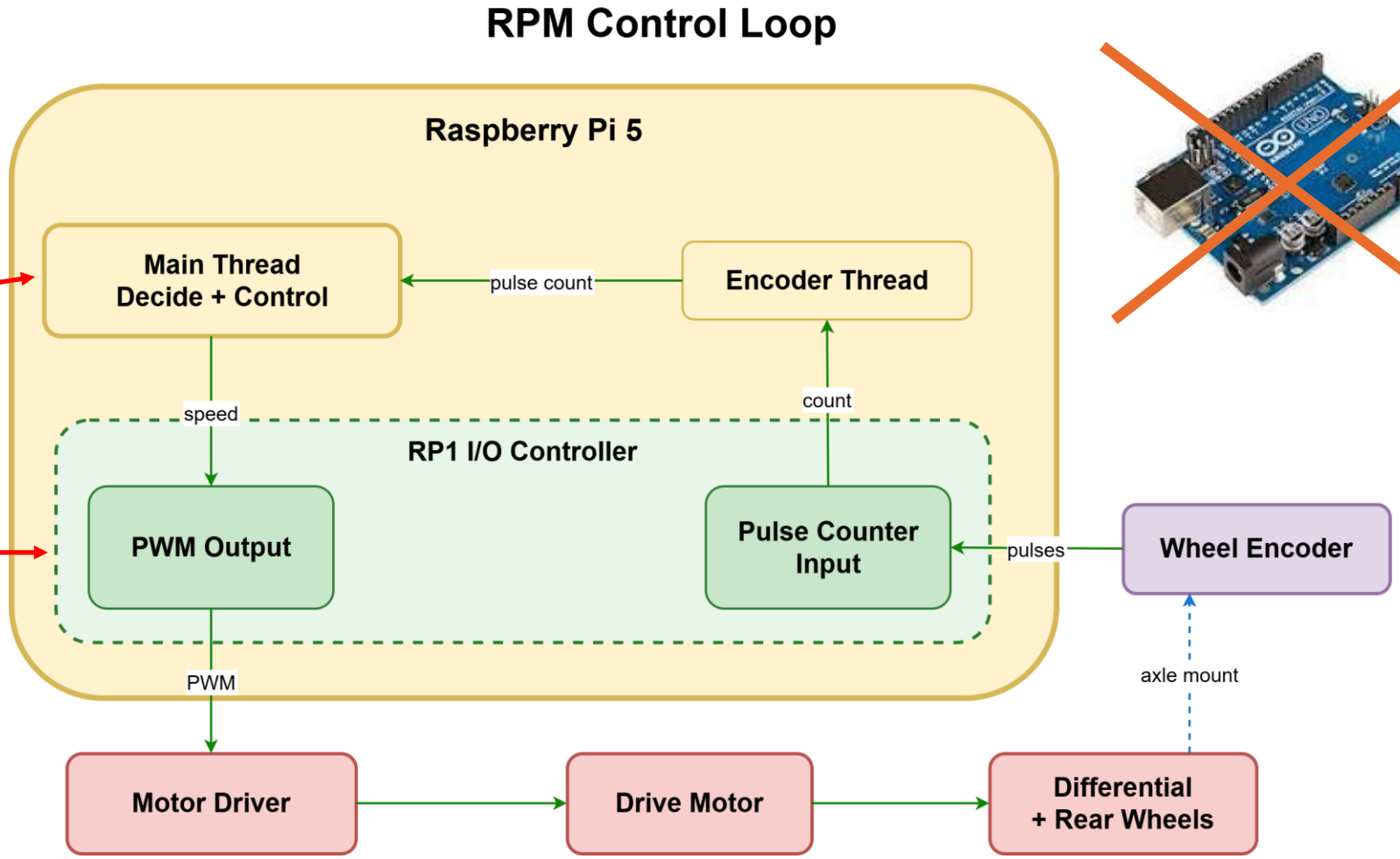
Greenbotics System Integration

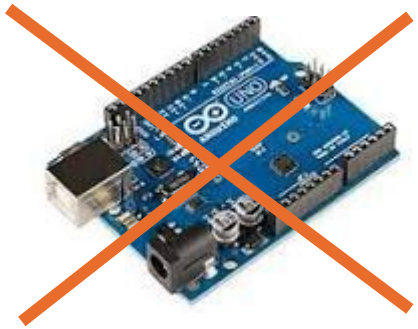


Cross-Subsystem Decision Case

One Raspberry Pi 5 instead of a dual-controller architecture

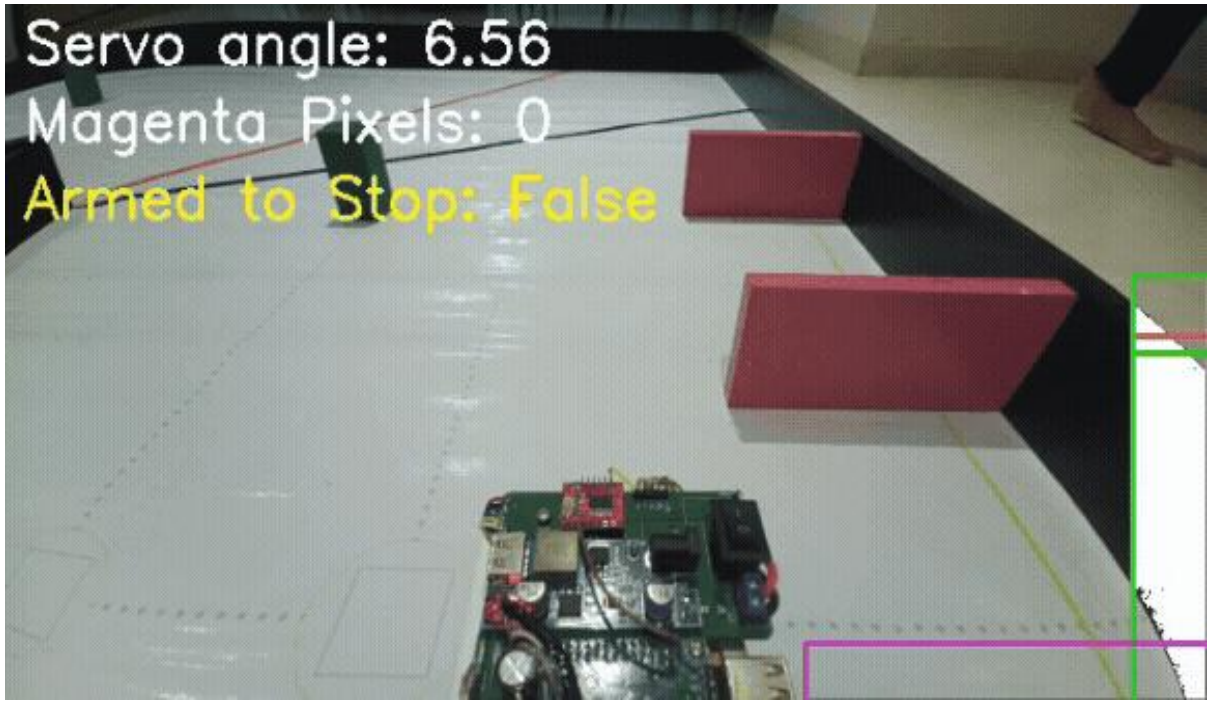




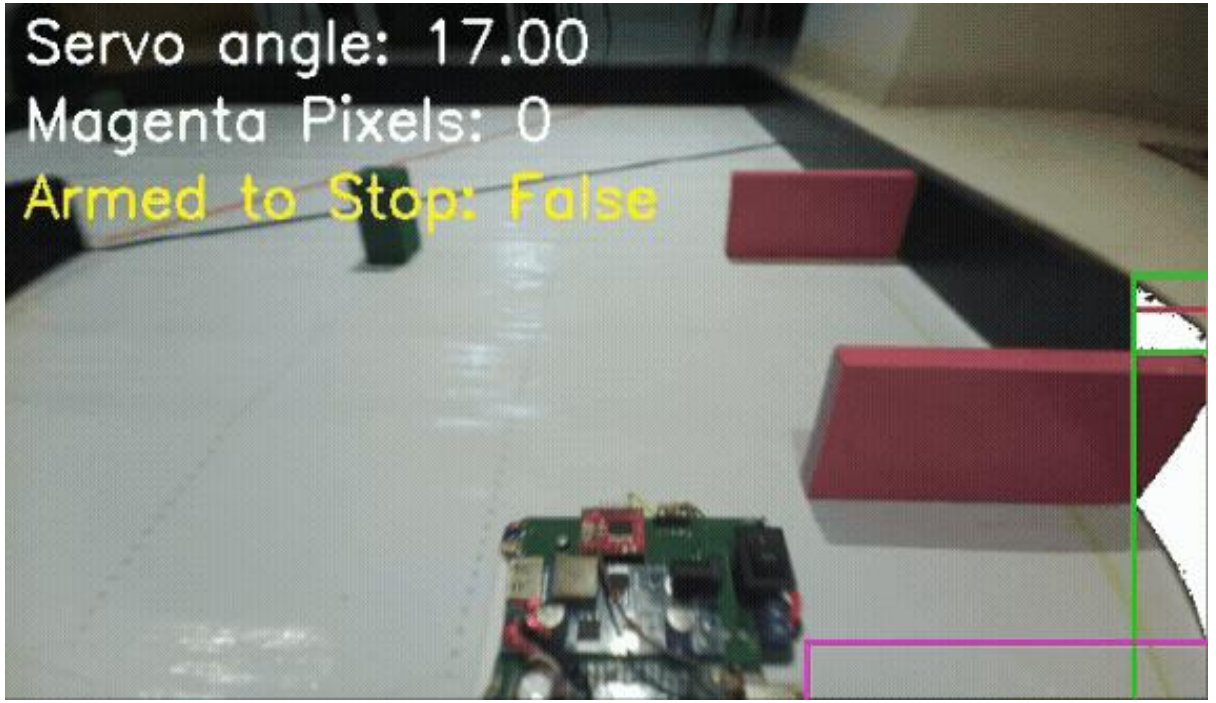


Failure- Wheel wobble showing up as camera jitter

BEFORE

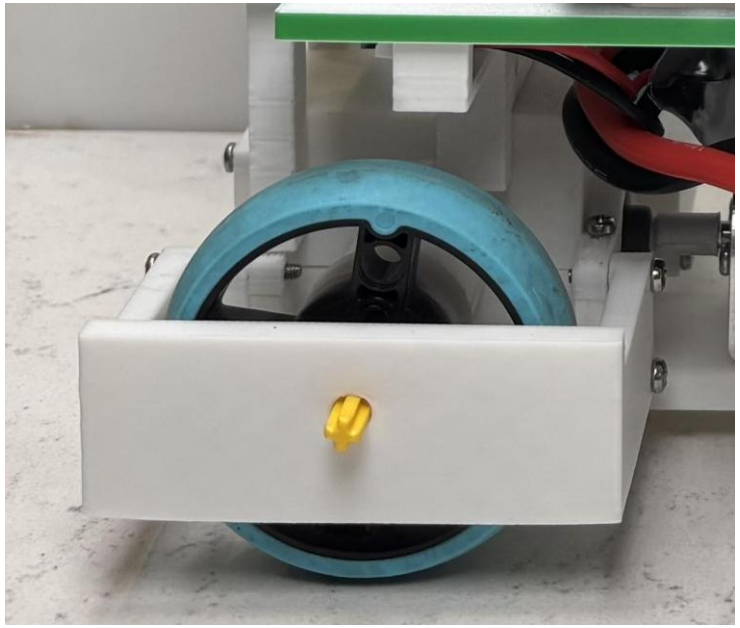


AFTER

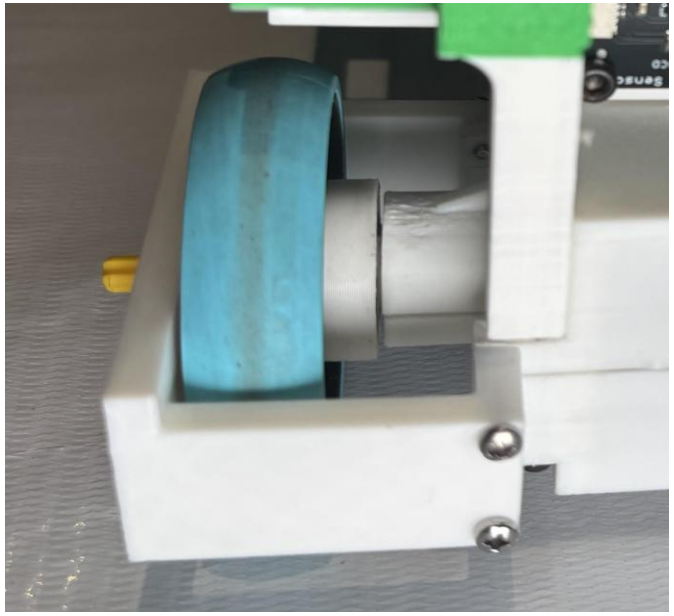


Reduce Wobble

Solution- Wheel Bracket as Stabilizer



Side View



Back View

Reproducibility & GitHub Quality



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Files

main

Go to file

- docs
- matlab
- models
- other
- schemes
- src
  - motors
  - obstacle_challenge
  - open_challenge
  - sensors
  - tools
  - README.md
  - __init__.py
- t-photos
- v-photos
- video
- .gitignore
- README.md

greenbotics-2026 / src /

Devansh-awat refactor: organize codebase into tools/ and legacy/ directories and a... e2210e2 · 3 days ago History

Name	Last commit message	Last commit date
..		
motors	refactor: organize codebase into tools/ and legacy/ directories and a...	3 days ago
obstacle_challenge	refactor: organize codebase into tools/ and legacy/ directories and a...	3 days ago
open_challenge	feat: implement modular obstacle challenge architecture with new visi...	3 days ago
sensors	refactor: organize codebase into tools/ and legacy/ directories and a...	3 days ago
tools	refactor: organize codebase into tools/ and legacy/ directories and a...	3 days ago
README.md	Enhance README with project overview and instructions	9 months ago
__init__.py	Added relative imports and logic to turn if in corner section	last year

README.md

Greenbotics Control Software (`src`)

This folder contains all Python code that runs on the Raspberry Pi 5 to control the Greenbotics robot for the WRO Future Engineers 2025 Open and Obstacle challenges. This code is split into smaller modules for sensors, motors and high-level challenge logic.

Layout

FROM INDIA TO PUERTO RICO... HERE WE COME!

